GLOWINSKI'S METHOD FOR VARIATIONAL QUASILINEAR EQUATIONS WITH NONSTANDARD GROWTH: REGULARITY, DISCRETIZATION AND THE LAVRENTIEV PHENOMENON

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Abstract. We study Glowinski's decomposition–coordination method (ALG2, an alternating direction method of multipliers) for minimising $\int_\Omega G(x,|\nabla v|) - fv$, where $G(x,\cdot)$ is a convex Young function: p-, $p(x)$-, Orlicz, double phase and two-phase energies. If the augmented Lagrangian has a saddle point (assumption (S)), the iteration converges weakly for every such G, and strongly under Lieberman's condition, with no continuity in x; without (S) the iterates are unbounded. We review when (S) holds and show that it differs from the absence of a Lavrentiev gap. For P1 and Crouzeix–Raviart discretisations the iteration converges for every penalty parameter, with no regularity. With a gap, P1 approximates the wrong infimum. For integrands constant on each triangle, $f = 0$ and edge-mean boundary data, we prove that the Crouzeix–Raviart energy is a guaranteed lower bound for the true infimum; with the P1 energy this gives a computable bracket whose width bounds the gap. Checkerboard experiments illustrate this.

1. Introduction

Let $\Omega \subset \mathbb{R}^d$ be a bounded open set and let $G\colon \Omega \times [0,\infty) \to [0,\infty)$ be such that $G(x,\cdot)$ is a convex Young function for a.e. x. We are interested in the numerical solution of

$$\min_v \; J(v), \qquad J(v) = \int_\Omega G(x,|\nabla v|)\,dx - \int_\Omega fv\,dx, \tag{1.1}$$

under Dirichlet conditions, whose Euler–Lagrange equation is the *g-Laplace equation*

$$-\operatorname{div}\Big(g(x,|\nabla u|)\frac{\nabla u}{|\nabla u|}\Big) = f \quad \text{in } \Omega, \qquad g = \partial_t G. \tag{1.2}$$

This class contains the p-Laplacian ($G = t^p/p$), the $p(x)$-Laplacian of electrorheological fluids [2], double phase and two-phase energies $t^p/p + a(x)t^q/q$, $a \geq 0$ [57, 25], Orlicz energies such as $t^p \log^a(1+t)$, and energies of exponential growth such as $e^t - 1 - t$, which violate the Δ_2 condition. Nonstandard growth brings two difficulties that are absent for the p-Laplacian with $p \geq 2$: the energy is not uniformly convex on L^2, and, for x-dependent G, smooth functions need not be dense in the energy space, so that the infimum of J over smooth (or Lipschitz) functions may be strictly larger than over all finite-energy functions. This is the Lavrentiev phenomenon [57, 58, 7].

The method. The decomposition–coordination method of Glowinski and Marrocco [44] and Fortin and Glowinski [40, 41, 43] (algorithm ALG2; with the step $\rho = r$ used here, an alternating direction method of multipliers, ADMM [45, 15]) introduces an auxiliary variable $\nu = \nabla u$ and a multiplier η, and alternates a *linear* Poisson-type problem for u (the same matrix at every iteration), a *pointwise* nonlinear problem for ν, which reduces to one scalar monotone equation $g(x,t)+rt = r|w|$ per point, and an explicit update of η, which at convergence is the flux $g(x,|\nabla u|)\nabla u/|\nabla u|$. The parameter $r > 0$ is the penalty parameter of the augmented Lagrangian. Since G enters only through the scalar equation, the code of [4] for the $p(x)$-Laplacian extends verbatim to arbitrary Young functions, and

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the conforming P1 discretisation can be replaced by a Crouzeix–Raviart (CR) one without touching the nonlinear step.

Two independent questions. Using such a method for (1.1) raises two questions that are easily confused:

(I) *Does the algorithm converge?* At the continuous level this hinges on Assumption (S): the augmented Lagrangian has a saddle point, i.e. J has a minimiser on H^1_0 whose flux lies in L^2 (Section 3.3 explains why this is not the same as the absence of a gap). At the discrete level the answer turns out to be *always yes*.

(II) *Does the discrete minimiser approximate the right object?* If there is no Lavrentiev gap, conforming P1 elements are adequate; if there is one, P1 energies converge to the infimum $m_{\mathcal{H}}$ over Lipschitz functions, not to the true infimum $m_{\mathcal{W}}$ ([21]; Proposition 5.1 for coefficients constant on the triangles), whereas CR energies converge to $m_{\mathcal{W}}$ under the assumptions of Balci, Ortner and Storn [8].

The paper follows these two questions in this order; each section raises the question answered by the next one.

Contributions. *What is classical.* The algorithm and its convergence theory under uniform convexity are Glowinski's [43, Ch. VI]. In particular the Lyapunov inequality behind our Theorem A is [43, Ch. VI, (5.25)] (our identity (2.9) is its equality form for $\rho = r$), the weak convergence of the whole multiplier sequence via Opial's lemma is [43, Ch. VI, §7], and the existence of discrete saddle points is [43, Ch. VI, §4.2]. Finite-dimensional ADMM convergence is [36, Thm. 8]. The convergence of CR energies to $m_{\mathcal{W}}$ is [8, Thm. 1], the generalised Marini formula and discrete strong duality for CR and Raviart–Thomas elements for convex problems are discussed by Bartels (see [11]) (see also [6, Prop. 2.37] and, for the linear case, [51, 5]); lower energy bounds from CR elements for problems without a gap are studied in [20] (not read by us).

What is new. (1) A convergence theory under (S) alone, for $\rho = r$ (Section 2): weak convergence for every convex Young integrand, including non-Δ_2 and nonsmooth ones (Theorem A); strong convergence for every Musielak–Orlicz integrand satisfying Lieberman's condition $\delta \le tg'/g \le g_0$ uniformly in x, with no continuity in x (Theorem B, Corollary 2.15); this covers the p-Laplacian with $1 < p < 2$, the $p(x)$-Laplacian with measurable p and $p^- < 2$, and two-phase energies $t^p/p + a(x)t^q/q$ with $p < 2$ and discontinuous a, none of which is covered by Glowinski's theorem (Proposition 2.9); and unboundedness of the iterates when (S) fails (Proposition 2.11, from [36, Thm. 7]). (2) A catalogue of when (S) is known to hold (Section 3, Table 1) and the precise relation between (S) and the Lavrentiev gap: (S) concerns the infimum over H^1_0, which lies between $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$ (Proposition 3.7); (S) can fail without a gap (Example 3.9) and, in a degenerate example, hold with a gap (Example 3.8). (3) Theorem C: for P1 and CR discretisations the iteration converges for every $r > 0$ and every superlinear convex integrand, with no regularity (Section 4). (4) Theorem D: building on the discrete CR/RT$_0$ duality of [11], we show that, for polygons triangulated so that $G(x,\cdot)$ is constant on each triangle, the dual value of the equilibrated Raviart–Thomas field produced at *every* step of the CR iteration is a guaranteed lower bound for the *continuous* infimum $m_{\mathcal{W}}$, also in the presence of a Lavrentiev gap; for $f = 0$ and exact edge-mean data the CR minimum itself is such a bound. Together with the P1 energy (an upper bound for $m_{\mathcal{H}}$) this yields a computable bracket $D(\sigma) \le m_{\mathcal{W}} \le m_{\mathcal{H}} \le J(v_h)$, whose width is a rigorous upper bound for the gap (Corollary 5.5); and a diagnostic based on the size of discrete saddle points (Theorem 5.7). (5) Experiments (Section 6): smooth problems with exact solutions, and Lavrentiev problems (Zhikov-type $p(x)$ and two-phase checkerboards, an Esposito–Leonetti–Mingione double phase), with guaranteed brackets on meshes with up to 1.6×10^6 unknowns and a comparison with [8].

What is not proved. (i) (S) is a regularity property of the solution, available from the literature only in some cases (Section 3); when it fails, the pair (η_n, ν_n) is unbounded (Proposition 2.11), but the behaviour of u_n is open. (ii) We prove no rate, neither for the iteration nor for the discretisation; all orders in Section 6 are observations. (iii) We treat only $\rho = r$. (iv) The bracket certifies an *upper* bound for the gap, never its positivity: no computable lower bound for $m_{\mathcal{H}}$ is available, and the presence of a gap is only inferred from the behaviour of the width under refinement. (v) Theorem D needs $f = 0$ for the statement on the CR energy itself, meshes resolving the coefficients and exact edge means of the boundary datum. (vi) Sections 4–5 are written for $d = 2$.

Notation. $(\cdot,\cdot)$ and $\|\cdot\|$ denote the inner product and norm of $L^2(\Omega)$ or $L^2(\Omega;\mathbb{R}^d)$; other norms carry a subscript. For a convex $\Psi\colon X \to (-\infty, +\infty]$ on a Hilbert space, $\operatorname{dom}\Psi = \{\Psi < \infty\}$ and $\partial\Psi(q) = \{\zeta \in X : \Psi(q') \geq \Psi(q) + (\zeta, q' - q)\ \forall q' \in X\}$. For a convex function γ on $[0,\infty)$, $\gamma'_{\pm}$ are its one-sided derivatives and $\gamma^*(s) = \sup_{t\geq 0}(st - \gamma(t))$, $s \geq 0$, its conjugate. $\rightharpoonup$ denotes weak convergence. The boundary datum is always called ψ; $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$ denote the infima of J over finite-energy $W^{1,1}$ functions and over Lipschitz functions with the boundary datum ψ (Sections 3.3 and 5).

2. The method and its convergence: from the p-Laplacian to Musielak–Orlicz energies

2.1. Setting.

Definition 2.1 (Young integrand)**.** A function $G\colon \Omega \times [0,\infty) \to [0,\infty)$ is a *Young integrand* if $G(\cdot, t)$ is measurable for every $t \geq 0$ and, for a.e. $x \in \Omega$, $G(x,\cdot)$ is finite, convex and nondecreasing with $G(x,0) = 0$.

A finite convex nondecreasing function on $[0,\infty)$ vanishing at 0 is continuous, so G is a Carathéodory function. We write

$$\Phi(x,\xi) = G(x,|\xi|), \qquad g(x,t) = \partial_t^+ G(x,t) \quad (t \geq 0);$$

$\Phi(x,\cdot)$ is convex on $\mathbb{R}^d$ and $g(x,\cdot)$ is nondecreasing, finite and nonnegative, with $g(x,0) = \inf_{s>0} G(x,s)/s$. No growth condition ($\Delta_2$, ∇_2, superlinearity) is assumed in this section. Let $V = H^1_0(\Omega)$ with $\|v\|_V = \|\nabla v\|$, $H = L^2(\Omega;\mathbb{R}^d)$, $f \in L^2(\Omega)$, and

$$F(q) = \int_\Omega \Phi(x, q(x))\,dx \in [0,+\infty] \quad (q \in H), \qquad J(v) = F(\nabla v) - (f,v) \quad (v \in V).$$

F is convex, $F \geq 0 = F(0)$, and F is lower semicontinuous on H (Fatou's lemma along a.e. convergent subsequences), hence weakly lower semicontinuous.
Abstract structure. We write $B = \nabla\colon V \to H$ and $\ell(v) = (f,v)$. Sections 2.2–2.5 use only:

(A1) V and H are real Hilbert spaces and $B\colon V \to H$ is linear with $\|Bv\|_H = \|v\|_V$;
(A2) $F\colon H \to [0,+\infty]$ is convex, lower semicontinuous and proper;
(A3) $\ell \in V^*$.

Section 4 uses the same structure for the finite element problems.

2.2. Augmented Lagrangian and saddle points.

For $r > 0$ define on $V \times H \times H$

$$\mathcal{L}_r(v,q,\mu) = F(q) - \ell(v) + (\mu, Bv - q) + \frac{r}{2}\|Bv - q\|^2.$$

A triple (u^*,ν^*,η^*) is a saddle point if $\mathcal{L}_r(u^*,\nu^*,\mu) \leq \mathcal{L}_r(u^*,\nu^*,\eta^*) \leq \mathcal{L}_r(v,q,\eta^*)$ for all (v,q,μ), with the middle term finite.

Proposition 2.2 (Saddle points)**.** *Assume (A1)–(A3). (u^*, ν^*, η^*) is a saddle point of $\mathcal{L}_r$ if and only if*

$$\nu^* = Bu^*, \qquad \eta^* \in \partial F(\nu^*), \qquad (\eta^*, Bw) = \ell(w) \quad \forall w \in V. \tag{2.1}$$

In that case $F(\nu^*) < \infty$ and u^* minimises $J(v) = F(Bv) - \ell(v)$ over V. The set of saddle points does not depend on r.

Proof. ($\Leftarrow$) $\mathcal{L}_r(u^*, \nu^*, \mu) = F(\nu^*) - \ell(u^*)$ does not depend on μ and is finite. For (v, q), the subgradient inequality $F(q) \geq F(\nu^*) + (\eta^*, q - \nu^*)$ and (2.1) with $w = v - u^*$ give $\mathcal{L}_r(v, q, \eta^*) - \mathcal{L}_r(u^*, \nu^*, \eta^*) \geq (\eta^*, B(v - u^*)) - \ell(v - u^*) + \frac{r}{2}\|Bv - q\|^2 \geq 0$.

($\Rightarrow$) The middle value is finite, so $F(\nu^*) < \infty$. The left inequality for all μ forces $Bu^* = \nu^*$. The right inequality with $(v, q) = (u^* + sw, \nu^*)$ reads $s[(\eta^*, Bw) - \ell(w)] + \frac{rs^2}{2}\|Bw\|^2 \geq 0$ for $s \in \mathbb{R}$, whence $(\eta^*, Bw) = \ell(w)$. With $v = u^*$ and $q = \nu^* + s(q' - \nu^*)$, $q' \in \operatorname{dom} F$, $s \in (0, 1)$, convexity gives $0 \leq s(F(q') - F(\nu^*)) - s(\eta^*, q' - \nu^*) + \frac{rs^2}{2}\|q' - \nu^*\|^2$; dividing by s and letting $s \downarrow 0$ gives $\eta^* \in \partial F(\nu^*)$. Finally, if (2.1) holds, $J(v) - J(u^*) \geq (\eta^*, B(v - u^*)) - \ell(v - u^*) = 0$. □

Assumption S. The augmented Lagrangian $\mathcal{L}_r$ has a saddle point in $V \times H \times H$; equivalently, (2.1) has a solution.

In the concrete case, (S) says that J has a minimiser $u \in H^1_0(\Omega)$ whose *flux* $\eta \in \partial F(\nabla u)$ belongs to $L^2(\Omega; \mathbb{R}^d)$ and satisfies $(\eta, \nabla w) = (f, w)$ for all $w \in H^1_0$. It is a property of the solution, not only of G (Section 3).

Lemma 2.3 (Pointwise description of ∂F)**.** *Let $q \in \operatorname{dom} F$ and $\eta \in H$. Then $\eta \in \partial F(q)$ if and only if $\eta(x) \in \partial\Phi(x, \cdot)(q(x))$ for a.e. $x \in \Omega$.*

Proof. ($\Leftarrow$) Integrate the pointwise inequality when $q' \in \operatorname{dom} F$. ($\Rightarrow$) Fix $z \in \mathbb{Q}^d$, $M \in \mathbb{N}$, $\Omega^z_M = \{x : \Phi(x, z) \leq M\}$ and a measurable $A \subset \Omega^z_M$. Testing with $q' = q\mathbf{1}_{\Omega\setminus A} + z\mathbf{1}_A \in \operatorname{dom} F$ gives $\int_A [\Phi(x, z) - \Phi(x, q) - \eta \cdot (z - q)] \geq 0$, so the integrand is ≥ 0 a.e. on Ω^z_M, hence a.e. on Ω. Taking the union of the null sets over $z \in \mathbb{Q}^d$ and using the continuity of $\Phi(x, \cdot)$ gives the claim. □

Lemma 2.4 (Subdifferential of Φ)**.** *Let $\gamma\colon [0, \infty) \to [0, \infty)$ be finite, convex, nondecreasing, $\gamma(0) = 0$, and $\phi(\xi) = \gamma(|\xi|)$.* (i) *$\partial\phi(0) = \{\zeta : |\zeta| \leq \gamma'_+(0)\}$.* (ii) *If $\xi \neq 0$ and γ is differentiable at $|\xi|$, then $\partial\phi(\xi) = \{\gamma'(|\xi|)\xi/|\xi|\}$.* (iii) *If $\gamma \in C^1([0, \infty))$ with $\gamma'(0) = 0$, then $\phi \in C^1(\mathbb{R}^d)$ and $\partial\phi(\xi) = \{T(\xi)\}$, $T(\xi) := \gamma'(|\xi|)\xi/|\xi|$, $T(0) = 0$.*

Proof. (i) $\zeta \in \partial\phi(0)$ iff $\gamma(t) \geq |\zeta| t$ for all $t \geq 0$, iff $|\zeta| \leq \inf_{t>0} \gamma(t)/t = \gamma'_+(0)$. (ii)–(iii): a convex function differentiable at ξ has $\partial\phi(\xi) = \{\nabla\phi(\xi)\}$, and $\nabla\phi(\xi) = \gamma'(|\xi|)\xi/|\xi|$ for $\xi \neq 0$; in (iii), $\phi(\xi) = o(|\xi|)$ at 0. □

Hence, if $G(x, \cdot) \in C^1([0, \infty))$ with $g(x, 0) = 0$, then for $q \in \operatorname{dom} F$

$$\partial F(q) = \{T(\cdot, q)\} \ \text{ if } T(\cdot, q) \in H, \qquad \partial F(q) = \emptyset \ \text{ otherwise;} \qquad T(x, \xi) = g(x, |\xi|)\frac{\xi}{|\xi|}. \tag{2.2}$$

For $G = e^t - 1$, instead, $g(x, 0) = 1$ and ∂F is multivalued at fields that vanish on a set of positive measure.

2.3. **The algorithm.** Fix $r > 0$, $\nu_0 \in H$ and $\eta_1 \in H$ (the code uses $\nu_0 = \eta_1 = 0$). For $n \geq 1$:

$$u_n \in V : \quad (\eta_n + r(Bu_n - \nu_{n-1}), Bw) = \ell(w) \quad \forall w \in V, \tag{2.3}$$

$$\nu_n = \operatorname*{argmin}_{q \in H} \Big\{ F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2 \Big\}, \tag{2.4}$$

$$\eta_{n+1} = \eta_n + r(Bu_n - \nu_n). \tag{2.5}$$

Step (2.3) minimises $\mathcal{L}_r(\cdot, \nu_{n-1}, \eta_n)$, step (2.4) minimises $\mathcal{L}_r(u_n, \cdot, \eta_n)$, and (2.5) is a multiplier step of length $\rho = r$. This is ALG2 of [41], [43, Ch. VI, (3.7)–(3.10)] with $\rho = r$ and the same order of the

steps, as used in [4]. In the concrete case (2.3) is the Poisson problem $-r\Delta u_n = f + \operatorname{div}(\eta_n - r\nu_{n-1})$, $u_n \in H^1_0$. We write $s_n = Bu_n - \nu_n$ (primal residual), so that $\eta_{n+1} - \eta_n = r s_n$.

Lemma 2.5 (Well-posedness)**.** *Assume (A1)–(A3).* (i) (2.3) *has a unique solution.* (ii) (2.4) *has a unique solution ν, characterised by $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$; consequently*

$$\eta_{n+1} \in \partial F(\nu_n) \qquad (n \geq 1). \tag{2.6}$$

(iii) *For $n \geq 1$,*

$$(\eta_{n+1} + r(\nu_n - \nu_{n-1}), Bw) = \ell(w) \qquad \forall w \in V. \tag{2.7}$$

Proof. (i) Riesz' theorem in V with the inner product $r(Bv, Bw)$. (ii) $\Psi_n(q) = F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2$ is proper, convex, l.s.c., coercive and strictly convex, so it has a unique minimiser; the variational characterisation follows by testing with $\nu + s(q - \nu)$, $s \downarrow 0$, and conversely from the subgradient inequality and $\frac{r}{2}\|Bu_n - q\|^2 - \frac{r}{2}\|Bu_n - \nu\|^2 = -r(Bu_n - \nu, q - \nu) + \frac{r}{2}\|q - \nu\|^2$. (iii) Insert $\eta_n + r(Bu_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1})$ into (2.3). □

Lemma 2.6 (The pointwise problem)**.** *Let γ and ϕ be as in Lemma 2.4, $r > 0$, $w \in \mathbb{R}^d$.*

(i) *$\xi \mapsto \phi(\xi) + \frac{r}{2}|\xi - w|^2$ has a unique minimiser $\hat\xi = \hat t\, w/|w|$ ($\hat\xi = 0$ if $w = 0$), where $\hat t \in [0, |w|]$ minimises $\chi(t) = \gamma(t) + \frac{r}{2}(t - |w|)^2$ on $[0, \infty)$.*

(ii) *$\hat t = 0$ if and only if $r|w| \leq \gamma'_+(0)$; otherwise $\gamma'_-(\hat t) \leq r(|w| - \hat t) \leq \gamma'_+(\hat t)$. If $\gamma \in C^1$ and $\gamma'(0) < r|w|$, $\hat t$ is the unique root in $(0, |w|]$ of the increasing function $t \mapsto \gamma'(t) + rt - r|w|$.*

(iii) *$\hat\eta := r(w - \hat\xi) \in \partial\phi(\hat\xi)$ and $|\hat\eta| = r(|w| - \hat t)$.*

Proof. (i) χ is continuous, strictly convex and coercive, and $\chi(t) > \chi(|w|)$ for $t > |w|$. Since $|\xi - w| \geq ||\xi| - |w||$, $\phi(\xi) + \frac{r}{2}|\xi - w|^2 \geq \chi(|\xi|) \geq \chi(\hat t)$, with equality at $\hat\xi$. (ii) is the optimality condition $\chi'_+(\hat t) \geq 0 \geq \chi'_-(\hat t)$. (iii) If $\hat t = 0$ use Lemma 2.4(i). If $\hat t > 0$, let $\hat w = w/|w|$ and $\lambda = r(|w| - \hat t) \in [\gamma'_-(\hat t), \gamma'_+(\hat t)]$; for every z, by monotonicity and convexity of γ (and $\gamma(0) = 0 \geq \gamma(\hat t) - \lambda\hat t$ when $\hat w \cdot z < 0$), $\phi(z) \geq \gamma(\max\{\hat w \cdot z, 0\}) \geq \gamma(\hat t) + \lambda(\hat w \cdot z - \hat t) = \phi(\hat\xi) + \hat\eta \cdot (z - \hat\xi)$. □

Corollary 2.7 (Pointwise form of (2.4))**.** *In the concrete case, let $w_n = \eta_n/r + \nabla u_n$. For a.e. x, $\nu_n(x) = t(x)\, w_n(x)/|w_n(x)|$, where $t(x) = 0$ if $r|w_n(x)| \leq g(x, 0)$ and otherwise, when $G(x, \cdot) \in C^1$, $t(x) \in (0, |w_n(x)|]$ solves*

$$g(x, t) + r\, t = r|w_n(x)|. \tag{2.8}$$

Moreover $\eta_{n+1}(x) = r(w_n(x) - \nu_n(x)) \in \partial\Phi(x, \cdot)(\nu_n(x))$: the multiplier is the flux.

Proof. By (2.6) and Lemma 2.3, $\eta_{n+1}(x) = r(w_n(x) - \nu_n(x)) \in \partial\Phi(x, \cdot)(\nu_n(x))$ a.e., so $\nu_n(x)$ minimises $\Phi(x, \cdot) + \frac{r}{2}|\cdot - w_n(x)|^2$; apply Lemma 2.6. □

2.4. **What Glowinski's theorem covers: from $p \geq 2$ to nonstandard growth.** Glowinski [43, Ch. VI, §§4–5] considers $\min_{v \in V} F(Bv) + \Gamma(v)$ with F, Γ proper, convex and l.s.c., and assumes

(i) the augmented Lagrangian has a saddle point;
(ii) B is injective and $B(V)$ is closed in H;
(iii) $F(q)/\|q\| \to +\infty$ as $\|q\| \to \infty$;
(iv) $F = F_0 + F_1$ with F_0, F_1 proper, convex, l.s.c., F_0 Gâteaux differentiable and uniformly convex on bounded sets in the sense of [43, (4.6)]: for every M there is a continuous strictly increasing δ_M, $\delta_M(0) = 0$, with $(F'_0(q) - F'_0(p), q - p) \geq \delta_M(\|q - p\|)$ for $\|p\|, \|q\| \leq M$.

Then [43, Thm. 5.1], for ALG2 with $0 < \rho < \frac{1+\sqrt5}{2} r$: $u_n \to u$ strongly in V, $\nu_n \to Bu$ strongly in H, and (η_n) is bounded with multipliers as weak cluster points; by [43, §7] the whole sequence (η_n) converges weakly. For linear Γ, as here, the range $0 < \rho < 2r$ is recorded in [43, Rem. 5.1, §6.3]. In [43, §6.3] the s-Laplacian is posed in $W^{1,s}_0 \times L^s$, where the theorem "cannot be applied" for $s \neq 2$, and is treated after discretisation. On $V = H^1_0$, $H = L^2$, hypothesis (ii) is trivial, and (iii)–(iv) hold for a large class of integrands:

Lemma 2.8 (Glowinski's hypotheses on $H = L^2$)**.** *Let G be a Young integrand.*

(a) *If $t \mapsto G(x,t) - \frac{\varepsilon}{2}t^2$ is convex for some $\varepsilon > 0$ and a.e. x, then $F_0 = \frac{\varepsilon}{2}\|\cdot\|^2$, $F_1 = F - F_0$ satisfy* (iii)–(iv).

(b) *Let $G(x,\cdot) \in C^1$, $g(x,0) = 0$, $k(x,t) = g(x,t)/t$ nondecreasing in t, $k_*(t) := \operatorname{ess\,inf}_x k(x,t) > 0$ for $t > 0$ and $K := \operatorname{ess\,sup}_x k(x,1) < \infty$. Let $h(x,t) = G(x,t)$ for $t \le 1$ and $h(x,t) = G(x,1) + \frac{1}{2}k(x,1)(t^2-1)$ for $t > 1$. Then $F_0 = \int_\Omega h(x,|q|)\,dx$ and $F_1 = F - F_0$ satisfy* (iii)–(iv).

Proof. (a) $G(x,\cdot) - \frac{\varepsilon}{2}t^2$ is convex with nonnegative right derivative at 0, hence nondecreasing, so F_1 is convex, proper and l.s.c.; $F \ge F_0$ gives (iii). (b) $\partial_t h(x,t) = t\,k(x,\min\{t,1\}) \le Kt$, so F_0 is C^1 on H with Lipschitz gradient $q \mapsto k(\cdot,\min\{|q|,1\})q$; $G-h$ is convex and nondecreasing because $g(x,t)-g(x,s) \ge (t-s)k(x,1)$ for $t > s \ge 1$. Lemma 2.13 below, applied to $k(x,\min\{t,1\})$, and the splitting argument of the proof of Theorem B(b) give $(F_0'(q)-F_0'(p), q-p) \ge \frac{3}{8}k_*(\min\{\frac{\|q-p\|}{4|\Omega|^{1/2}},1\})\|q-p\|^2$, and replacing k_* by the continuous minorant $\tau^{-1}\int_0^\tau k_*$ yields an admissible δ_M. Finally $G \ge h \ge \frac{1}{2}k_*(1)(t^2-1)$ gives (iii). □

Lemma 2.8 covers t^p/p and $p(x)$ with $2 \le p(x) \le p^+ < \infty$ (part (b)), which justifies [4, Thm. 4.3], and uniformly convex integrands such as $e^t - 1$, $e^t - 1 - t$, $t^2/2 + a(x)t^4/4$ with measurable $a \ge 0$, $t^{3/2}/(3/2) + t^3/3$ (part (a)). For these, under (S), Glowinski's theorem gives strong convergence. Below $p = 2$ it gives nothing:

Proposition 2.9. *Let $F(q) = \frac{1}{p}\int_\Omega |q|^p\,dx$ on $H = L^2(\Omega;\mathbb{R}^d)$ with $1 < p < 2$. Then F satisfies neither* (iii) *nor* (iv).

Proof. Fix a unit vector e and, for large M, $A_M \subset \Omega$ with $|A_M| = M^{-2+\varepsilon}$, $0 < \varepsilon < 2(2-p)$; $q_M = M\mathbf{1}_{A_M}e$ has $\|q_M\| = M^{\varepsilon/2} \to \infty$ and $F(q_M)/\|q_M\| = M^{p-2+\varepsilon/2}/p \to 0$: (iii) fails. With $|A_M| = M^{-2}$, $\|q_M\| = 1$ and $\frac{1}{2}(F(0) + F(q_M)) - F(\frac{q_M}{2}) = \frac{M^{p-2}}{p}(\frac{1}{2} - 2^{-p}) \to 0$. If $F = F_0 + F_1$ as in (iv), then with $m = q_M/2$ and $q \in \{0, q_M\}$, $F_0(q) - F_0(m) - (F_0'(m), q-m) = \int_0^1 (F_0'(m + s(q-m)) - F_0'(m), q-m)\,ds \ge \int_0^1 \delta_1(s/2)\frac{ds}{s} \ge \frac{1}{2}\delta_1(\frac{1}{4})$; averaging over q, $\frac{1}{2}(F_0(0) + F_0(q_M)) - F_0(m) \ge \frac{1}{2}\delta_1(\frac{1}{4}) > 0$, while the same expression for F_1 is ≥ 0: a contradiction. □

The proof localises: for $G = t^{p(x)}/p(x)$ with $|\{p \le p_0\}| > 0$ for some $p_0 < 2$, choosing $A_M \subset \{p \le p_0\}$ gives the same conclusions. Thus the classical theorem covers neither the p-Laplacian with $p < 2$ nor variable exponents with $p^- < 2$. On the other hand, nothing in the *algorithm* requires Δ_2 or ∇_2: the steps (2.3)–(2.5) are defined for every Young integrand. The next two subsections give a theory that matches this generality: weak convergence under (S) alone, for every convex Young integrand (Theorem A), and strong convergence for every Musielak–Orlicz integrand with Lieberman's structure, uniformly in x but without any continuity in x (Theorem B).

2.5. **Theorem A: weak convergence under (S).**

Theorem A. *Assume (A1)–(A3) and Assumption S. Let (u_n, ν_n, η_n) be generated by (2.3)–(2.5), let (u^*, ν^*, η^*) be any saddle point, and set*

$$E_n = \frac{1}{2r}\|\eta_{n+1} - \eta^*\|^2 + \frac{r}{2}\|\nu_n - \nu^*\|^2 \qquad (n \ge 0).$$

Then for every $n \ge 1$

$$E_n + \frac{r}{2}\|s_n\|^2 + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2 + (\eta_{n+1} - \eta^*, \nu_n - \nu^*) + r(s_n, \nu_n - \nu_{n-1}) = E_{n-1}, \tag{2.9}$$

and the last two terms on the left are nonnegative (the last one for $n \geq 2$, and for $n = 1$ if $\eta_1 \in \partial F(\nu_0)$). Consequently (E_n) is nonincreasing; (η_n), (ν_n) are bounded in H and (u_n) in V;

$$\sum_{n\geq 1}\|s_n\|^2 < \infty, \qquad \sum_{n\geq 1}\|\nu_n - \nu_{n-1}\|^2 < \infty, \qquad \sum_{n\geq 1} m_n < \infty, \quad m_n := (\eta_{n+1} - \eta^*, \nu_n - \nu^*) \geq 0.$$

Moreover there is a saddle point $(\bar u, \bar\nu, \bar\eta)$ with $u_n \rightharpoonup \bar u$ in V, $\nu_n \rightharpoonup \bar\nu$ and $\eta_n \rightharpoonup \bar\eta$ in H. In the concrete case $u_n \to \bar u$ strongly in $L^2(\Omega)$, $\nabla u_n - \nu_n \to 0$ in L^2, and $\bar u$ minimises J over $H^1_0(\Omega)$.

No growth condition and no differentiability of G is used: whenever (S) holds, Theorem A applies to t^p/p for every $p > 1$, to $p(x)$, double phase and two-phase energies, to $t^p \log^a(1+t)$, to $e^t - 1 - t$ and to $e^t - 1$.

Proof. Step 1: the identity. Subtract $(\eta^*, Bw) = \ell(w)$ from (2.7) and take $w = u_n - u^*$, so that $Bw = s_n + (\nu_n - \nu^*)$:

$$(\eta_{n+1} - \eta^*, s_n) + (\eta_{n+1} - \eta^*, \nu_n - \nu^*) + r(\nu_n - \nu_{n-1}, s_n) + r(\nu_n - \nu_{n-1}, \nu_n - \nu^*) = 0.$$

By $2(a-c, a-b) = \|a-c\|^2 + \|a-b\|^2 - \|b-c\|^2$ and $rs_n = \eta_{n+1} - \eta_n$, the first term equals $\frac{1}{2r}(\|\eta_{n+1}-\eta^*\|^2 - \|\eta_n - \eta^*\|^2) + \frac{r}{2}\|s_n\|^2$ and the last one $\frac{r}{2}(\|\nu_n - \nu^*\|^2 - \|\nu_{n-1} - \nu^*\|^2) + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2$; this is (2.9).

Step 2: signs. By (2.6), (2.1) and the monotonicity of ∂F, $m_n \geq 0$; and $\eta_n \in \partial F(\nu_{n-1})$ gives $r(s_n, \nu_n - \nu_{n-1}) = (\eta_{n+1} - \eta_n, \nu_n - \nu_{n-1}) \geq 0$. Summing (2.9) gives the bounds and the summability; $Bu_n = s_n + \nu_n$ is bounded.

Step 3: weak cluster points are saddle points. Let $\nu_{n_k} \rightharpoonup \bar\nu$, $\eta_{n_k+1} \rightharpoonup \bar\eta$, $u_{n_k} \rightharpoonup \bar u$. Since $s_n \to 0$, $B\bar u = \bar\nu$; passing to the limit in (2.7), $(\bar\eta, Bw) = \ell(w)$. Taking $w = u_n$ in (2.7), $(\eta_{n+1}, \nu_n) = \ell(u_n) - (\eta_{n+1}, s_n) - r(\nu_n - \nu_{n-1}, Bu_n) \to \ell(\bar u) = (\bar\eta, \bar\nu)$ along n_k. By (2.6), $F(q) \geq F(\nu_{n_k}) + (\eta_{n_k+1}, q) - (\eta_{n_k+1}, \nu_{n_k})$; weak lower semicontinuity gives $F(q) \geq F(\bar\nu) + (\bar\eta, q - \bar\nu)$, so $\bar\eta \in \partial F(\bar\nu)$ and $(\bar u, \bar\nu, \bar\eta)$ is a saddle point (Proposition 2.2).

Step 4: the whole sequence (Opial's argument). Let $x_n = (\nu_n, \eta_{n+1})$ and $\langle(\nu,\eta),(\nu',\eta')\rangle = r(\nu,\nu') + \frac{1}{r}(\eta,\eta')$. For every saddle pair z, $|x_n - z|^2 = 2E_n(z)$ converges. If z_1, z_2 are weak cluster points, they are saddle pairs by Step 3, so $\langle x_n, z_1 - z_2\rangle$ converges, and evaluating the limit along the two subsequences gives $|z_1 - z_2| = 0$. Hence $x_n \rightharpoonup (\bar\nu, \bar\eta)$, $\eta_n \rightharpoonup \bar\eta$, and, B being injective, $u_n \rightharpoonup \bar u$. In the concrete case Rellich's theorem gives $u_n \to \bar u$ in L^2. □

Remark 2.10 (Initialisation; related results). Since $s_n + (\nu_n - \nu_{n-1}) = Bu_n - \nu_{n-1}$, (2.9) reads $E_n + m_n + \frac{r}{2}\|Bu_n - \nu_{n-1}\|^2 = E_{n-1}$ for every $n \geq 1$, whatever η_1 is; so all conclusions hold for arbitrary $\nu_0, \eta_1 \in H$. Identity (2.9) is the equality version, for $\rho = r$, of Glowinski's inequality [43, Ch. VI, (5.25)], and Step 4 is the argument of [43, Ch. VI, §7]; Theorem A shows that (iii)–(iv) are not needed for weak convergence. The same functional appears in [12, Thm. 3.7] and [15, App. A]. Weak convergence also follows from the equivalence of ADMM with Douglas–Rachford splitting for the dual [42], [36, §5], and the weak convergence of the latter [49, 55, 13]; see [52]. Through the correspondence in the proof of Proposition 2.11, the example of Bùi and Combettes [18], in which the Douglas–Rachford shadow sequence converges weakly but not strongly, is an instance of (A1)–(A3) ($V = \mathbb{R}$, $Bt = te_0$, $\ell = 0$) in which that sequence is η_n; so the weak convergence of η_n cannot be improved under (A1)–(A3). The example says nothing about ν_n or u_n.

Proposition 2.11 ((S) is necessary for boundedness)**.** *Assume (A1)–(A3) and let (u_n, ν_n, η_n) be generated from any $\nu_0, \eta_1 \in H$. Then* (S) *holds if and only if $(\eta_n + r\nu_{n-1})_n$ is bounded in H. In particular, if* (S) *fails, (η_n) or (ν_n) is unbounded.*

Proof. If (S) holds, boundedness is Theorem A with Remark 2.10. Conversely, (2.3)–(2.5) is the ADMM of [36, Thm. 8] for $\min(-\ell(v)) + F(Bv)$ with penalty r, relaxation 1 and exact steps; the relations (Y1)–(Y3) of its proof are algebraic and hold in Hilbert space: $z_n := \eta_n + r\nu_{n-1}$ is the Douglas–Rachford sequence for the maximal monotone operators $\mathcal{A} = \partial\iota_{\{B^*\eta = \ell\}}$ and $\mathcal{B} = \partial F^*$.

The affine space $\{B^*\eta = \ell\}$ is nonempty and closed (B^* is onto by (A1)) with normal cone $R(B)$, and $0 \in \mathcal{A}\eta + \mathcal{B}\eta$ means $\eta \in \partial F(Bu)$ for some u with $B^*\eta = \ell$, i.e. (2.1). If (S) fails, $\operatorname{zer}(\mathcal{A}+\mathcal{B}) = \emptyset$ and (z_n) is unbounded by [36, Thm. 7], stated in a general Hilbert space. □

2.6. **Theorem B: strong convergence for Musielak–Orlicz integrands.** Assume now that $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$ for a.e. x, and let $T(x,\xi) = g(x,|\xi|)\xi/|\xi|$,

$$k(x,t) = \frac{g(x,t)}{t}\ (t>0), \qquad \varpi(x;a,b) = (T(x,a) - T(x,b))\cdot(a-b) \geq 0.$$

By (2.2) and (2.6), $\eta_{n+1} = T(\cdot,\nu_n)$ and $\eta^* = T(\cdot,\nu^*)$ for every saddle point, hence

$$m_n = \int_\Omega \varpi(x;\nu_n(x),\nu^*(x))\,dx \to 0. \tag{2.10}$$

Glowinski's hypothesis (iv) asks for a uniform lower bound of ϖ in terms of $|a-b|$ alone, which is false for $p<2$ (Proposition 2.9). What survives for nonstandard growth is a lower bound *up to a small multiple of the energy*:

Definition 2.12 (Modular uniform monotonicity)**.** G satisfies (UM) with *error modular* Θ if $\Theta\colon \Omega\times[0,\infty)\to[0,\infty)$ is Carathéodory, $\Theta(x,\cdot)$ is nondecreasing with $\Theta(x,t)>0$ for $t>0$, and there is $\omega\colon(0,1]\to[0,\infty)$ with $\omega(\varepsilon)\to 0$ as $\varepsilon\downarrow 0$ such that, for a.e. x, all $a,b\in\mathbb{R}^d$ and all $\varepsilon\in(0,1]$,

$$\Theta\big(x,|a-b|\big) \leq \frac{1}{\varepsilon}\,\varpi(x;a,b) + \omega(\varepsilon)\big(G(x,|a|)+G(x,|b|)\big). \tag{2.11}$$

Lemma 2.13. *If $t\mapsto k(x,t)$ is nondecreasing on $(0,\infty)$ and $k(x,0) := \lim_{t\downarrow 0} k(x,t)$, then $\varpi(x;a,b) \geq \frac12 k(x,\frac{|a-b|}{2})|a-b|^2$ for all a,b.*

Proof. With $k_a = k(x,|a|)$, $k_b = k(x,|b|)$, $\varpi = \frac12(k_a+k_b)|a-b|^2 + \frac12(k_a-k_b)(|a|^2-|b|^2)$, and the second term is ≥ 0. Hence $\varpi \geq \frac12 k(x,\max\{|a|,|b|\})|a-b|^2$, and $\max\{|a|,|b|\} \geq |a-b|/2$. □

Theorem B. *Assume* (S) *and that $G(x,\cdot)\in C^1([0,\infty))$ with $g(x,0)=0$ for a.e. x; let $(\bar u,\bar\nu,\bar\eta)$ be the limit in Theorem A.*

- (a) *If G satisfies* (UM)*, then $\int_\Omega \Theta(x,|\nu_n-\bar\nu|)\,dx\to 0$; $\nu_n\to\bar\nu$ and $\nabla u_n\to\nabla\bar u$ strongly in $L^s(\Omega;\mathbb{R}^d)$ for every $s\in[1,2)$, and $u_n\to\bar u$ in $W^{1,s}_0(\Omega)$. Moreover $\nu^*=\bar\nu$ for every saddle point, so the minimiser of J on $H^1_0(\Omega)$ is unique.*
- (b) *If $\Omega_+\subset\Omega$ is measurable, $k(x,\cdot)$ is nondecreasing for a.e. $x\in\Omega_+$ and $k_+(t) := \operatorname{ess\,inf}_{x\in\Omega_+} k(x,t) > 0$ for every $t>0$, then $\nu_n\to\bar\nu$ and $\nabla u_n\to\nabla\bar u$ in $L^2(\Omega_+;\mathbb{R}^d)$. If $\Omega_+=\Omega$, the minimiser of J on H^1_0 is unique.* (*No* (UM) *is needed in* (b).)

Proof. Fix a saddle point (u^*,ν^*,η^*) and put $e_n = \nu_n - \nu^*$. (a) By (2.6) and $F\geq 0 = F(0)$, $F(\nu_n)\leq(\eta_{n+1},\nu_n)\leq M := \sup_k\|\eta_{k+1}\|\sup_k\|\nu_k\| < \infty$, and $F(\nu^*)<\infty$. Integrating (2.11) with $a=\nu_n$, $b=\nu^*$ and using (2.10), $\int_\Omega\Theta(x,|e_n|)\leq m_n/\varepsilon + \omega(\varepsilon)(M+F(\nu^*))$ for $\varepsilon\in(0,1]$; letting $n\to\infty$ and then $\varepsilon\to 0$, $\int\Theta(x,|e_n|)\to 0$. Every subsequence has a further subsequence with $\Theta(x,|e_n(x)|)\to 0$, hence $e_n(x)\to 0$, a.e.; (e_n) is bounded in L^2, so $(|e_n|^s)$ is uniformly integrable for $s<2$ and Vitali's theorem gives $e_n\to 0$ in L^s for the whole sequence. As $\nu_n\rightharpoonup\bar\nu$, $\nu^*=\bar\nu$; then $\nabla u_n = s_n+\nu_n\to\nabla\bar u$ in L^s, and Poincaré's inequality gives $u_n\to\bar u$ in $W^{1,s}_0$. If u' minimises J on H^1_0, then $J(u')=J(u^*)$ and the subgradient inequality give $F(q)\geq F(\nabla u')+(\eta^*,q-\nabla u')$ for all q, i.e. $(u',\nabla u',\eta^*)$ is a saddle point; hence $\nabla u'=\bar\nu$ and $u'=\bar u$.

(b) On $\Omega_+\cap\{|e_n|>\delta\}$, Lemma 2.13 and the monotonicity of k give $\varpi(x;\nu_n,\nu^*)\geq\frac12 k_+(\delta/2)|e_n|^2$, so $\|e_n\|^2_{L^2(\Omega_+)}\leq\delta^2|\Omega| + 2m_n/k_+(\delta/2)$ and $e_n\to 0$ in $L^2(\Omega_+)$. Since $\nu_n\rightharpoonup\bar\nu$ also in $L^2(\Omega_+)$, $\nu^*=\bar\nu$ a.e. in Ω_+, and $\nabla u_n = s_n+\nu_n\to\nabla\bar u$ in $L^2(\Omega_+)$. If $\Omega_+=\Omega$, uniqueness follows as in (a). □

The key a priori fact in (a) is that, under (S), the energies $F(\nu_n)$ are bounded; (UM) then converts $m_n \to 0$ into convergence in the modular Θ. The next lemma shows that (UM) holds, with Θ comparable to G, for every integrand satisfying Lieberman's condition uniformly in x.

Lemma 2.14. *Let $G(x,\cdot) \in C^1([0,\infty)) \cap C^2(0,\infty)$, $g(x,0) = 0$, $g(x,t) > 0$ for $t > 0$, and assume that for some $0 < \delta \le g_0 < \infty$*

$$\text{(L)} \qquad \delta \le \frac{t\,\partial_t g(x,t)}{g(x,t)} \le g_0 \qquad \text{for a.e. } x \in \Omega \text{ and all } t > 0.$$

Put $\gamma = (g_0 - 1)_+$, $\beta = (1-\delta)_+$, $c_1 = \min\{1,\delta\}\,4^{-1-\gamma}$. Then for a.e. x and all $a, b \in \mathbb{R}^d$, with $S = |a| + |b|$:

(i) $\varpi(x;a,b) \ge c_1\, k(x,S)\,|a-b|^2$;

(ii) *if $\delta \ge 1$, $G(x,|a-b|) \le c_1^{-1}\varpi(x;a,b)$; if $\delta < 1$, for every $\varepsilon \in (0,1]$,*

$$G(x,|a-b|) \le \frac{1}{c_1\varepsilon}\,\varpi(x;a,b) + (1+g_0)2^{g_0}\varepsilon^{\kappa}\big(G(x,|a|) + G(x,|b|)\big), \qquad \kappa = \frac{1+\delta}{1-\delta};$$

in particular G satisfies (UM) *with $\Theta = c_1 G$;*

(iii) *if $g_0 < 1$, $|T(x,a) - T(x,b)| \le \frac{3}{\delta}\, g(x,|a-b|)$.*

Proof. Fix x and drop it. From (L), $(\log g)' \in [\delta/t, g_0/t]$ and $(\log k)' \in [(\delta-1)/t, (g_0-1)/t]$; integrating, for $0 < t \le S$,

(2.12)
$$k(S)\Big(\frac{t}{S}\Big)^{\gamma} \le k(t) \le k(S)\Big(\frac{S}{t}\Big)^{\beta}, \qquad g(2t) \le 2^{g_0} g(t), \quad G(2t) \le 2^{1+g_0} G(t), \quad tg(t) \le (1+g_0)G(t),$$

the last because $(tg)' \le (1+g_0)g$. (i) For $\xi \ne 0$, $DT(\xi)$ is symmetric with eigenvalues $k(|\xi|)$ and $g'(|\xi|) \in [\delta k(|\xi|), g_0 k(|\xi|)]$. With $e = a - b$, $c_s = b + se$, $s \mapsto T(c_s)$ is absolutely continuous (its derivative is bounded by $\max\{1,g_0\}k(|c_s|)|e|$, integrable since $k(t) \le k(1)t^{\delta-1}$ for $t \le 1$), so $\varpi(a,b) = \int_0^1 e \cdot DT(c_s) e\, ds \ge \min\{1,\delta\}|e|^2 \int_0^1 k(|c_s|)\, ds$. If $|a| \ge |b|$, then $|c_s| \ge S/4$ for $s \in [\frac34, 1]$, and (2.12) gives $\int_0^1 k(|c_s|)\,ds \ge \frac14 k(S) 4^{-\gamma}$; the other case is symmetric. (ii) $G(t) \le tg(t) = k(t)t^2$ and $|e| \le S$, so $G(|e|) \le k(S) S^{\beta}|e|^{2-\beta}$ by (2.12). If $\delta \ge 1$, $\beta = 0$ and (i) concludes. If $\delta < 1$, Young's inequality $X^{\theta}Y^{1-\theta} \le X + Y$ with $\theta = \frac{2-\beta}{2}$, $X = |e|^2/\varepsilon$, $Y = \varepsilon^{\kappa} S^2$ gives $S^{\beta}|e|^{2-\beta} \le |e|^2/\varepsilon + \varepsilon^{\kappa} S^2$; by (i) and (2.12), $G(|e|) \le \frac{\varpi}{c_1\varepsilon} + \varepsilon^{\kappa}(1+g_0)G(S)$, and $G(S) \le 2^{g_0}(G(|a|) + G(|b|))$ by convexity. (iii) If $g_0 < 1$, k is nonincreasing and $|DT| \le k$. Let s^* minimise $|c_s|$ over $\mathbb{R}$; then $|c_s| \ge |e||s - s^*|$ and $k(|c_s|) \le k(|e|)\max\{1, |s-s^*|^{\delta-1}\}$, so $|T(a) - T(b)| \le |e| \int_0^1 k(|c_s|)\,ds \le (1 + \frac{2}{\delta})g(|e|)$. □

Corollary 2.15 (Variable exponents, double phase, Lieberman-type integrands)**.** *Assume* (S) *and* (L)*, and let $(\bar u, \bar\nu, \bar\eta)$ be the limit in Theorem A.*

(i) $\int_\Omega G(x, |\nu_n - \bar\nu|)\,dx \to 0$ *(hence convergence in the Luxemburg norm of $L^{G(\cdot)}$, since* (L) *implies Δ_2 uniformly in x), and all conclusions of Theorem B(a) hold.*

(ii) *If $\delta \ge 1$ and $\operatorname{ess\,inf}_x g(x,1) > 0$, then $\nu_n \to \bar\nu$ and $\nabla u_n \to \nabla \bar u$ in L^2.*

(iii) *If $g_0 < 1$ and $K := \operatorname{ess\,sup}_x g(x,1) < \infty$, then $\eta_n \to \bar\eta$ strongly in L^2 and $\int_\Omega G(x, |\nabla u_n - \nabla \bar u|)\,dx \to 0$.*

Proof. (i) is Theorem B(a) with Lemma 2.14(ii). (ii) is Theorem B(b) with $\Omega_+ = \Omega$: $k(x,\cdot)$ is nondecreasing and $k(x,t) \ge g(x,1)\min\{1, t^{\gamma}\}$. (iii) By (2.12), $g(x,t) \le K(t^{\delta} + t^{g_0})$; by Lemma 2.14(iii), $|\eta_{n+1} - \bar\eta|^2 \le \frac{18K^2}{\delta^2}(|e_n|^{2\delta} + |e_n|^{2g_0})$, $e_n = \nu_n - \bar\nu$, which is uniformly integrable ($2g_0 < 2$) and tends to 0 a.e. along sub-subsequences; Vitali's theorem and $\eta_{n+1} - \eta_n \to 0$ give $\eta_n \to \bar\eta$ in L^2. Finally $G(x, |s_n + e_n|) \le 2^{g_0}(G(x,|s_n|) + G(x,|e_n|))$ with $G(x,t) \le 2K(t^{1+\delta} + t^2)$, $s_n \to 0$ in L^2, and (i). □

Examples. (L) holds, with *no continuity in* x, for $t^{p(x)}/p(x)$ with measurable p, $1 < p^- \le p \le p^+ < \infty$ ($\delta = p^- - 1$, $g_0 = p^+ - 1$; for constant $p \in (1,2)$ Corollary 2.15(i),(iii) gives strong convergence in $W_0^{1,p}$ and of η_n in L^2); for double phase and two-phase integrands $t^p/p + a(x)t^q/q$, $1 < p \le q < \infty$, with *any* measurable $a \ge 0$ ($\delta = p - 1$, $g_0 = q - 1$); and for $t^p \log^\alpha(e + t)$-type integrands. Thus, *whenever (S) holds*, the iteration converges strongly for the $p(x)$-Laplacian with $p^- < 2$ (including Zhikov's checkerboard exponent) and for two-phase problems with $p < 2$ and discontinuous coefficients, none of which is covered by Glowinski's theorem (for $p \ge 2$ Lemma 2.8 applies). For $e^t - 1 - t$, which violates (L) ($tg'/g \to \infty$), Theorem B(b) applies with $\Omega_+ = \Omega$ ($k = (e^t - 1)/t \ge 1$ is nondecreasing). For $e^t - 1$, where ∂F is multivalued, Glowinski's theorem applies via Lemma 2.8(a).

Remark 2.16 (Nonhomogeneous Dirichlet data). If $u = \psi$ on $\partial\Omega$ with $\psi \in H^1(\Omega)$ and $F(\nabla\psi) < \infty$, write $u = \psi + v$, $v \in H_0^1$, and $\tilde F(q) = F(q + \nabla\psi)$. Then $(V, H, \nabla, \tilde F, \ell)$ satisfies (A1)–(A3), and Proposition 2.2 and Theorems A–B apply in the variables (v, q, η) with $\nu = q + \nabla\psi$; in the proof of Theorem B(a), $F(\nu_n) = \tilde F(q_n) \le \tilde F(0) + (\eta_{n+1}, q_n)$ is still bounded.

3. When does (S) hold, and what does it say about the Lavrentiev gap?

Theorems A–B reduce the convergence of the continuous iteration to Assumption (S), and Proposition 2.11 shows that nothing weaker suffices. (S) is a regularity property of the solution. This section collects what is known about it, and then separates it from a different property, the absence of a Lavrentiev gap, which will govern the discrete problem in Section 5.

3.1. Sufficient conditions, and the p-Laplacian.

Lemma 3.1 (Lipschitz solutions). *Assume* $G(x,\cdot) \in C^1([0,\infty))$, $g(x,0) = 0$ *for a.e.* x, *and* $\operatorname{ess\,sup}_x g(x,M) < \infty$ *for every* $M > 0$. *If* $u \in H_0^1(\Omega)$ *has* $\nabla u \in L^\infty$ *and* $\int_\Omega T(x,\nabla u)\cdot\nabla\varphi = \int_\Omega f\varphi$ *for all* $\varphi \in C_c^\infty(\Omega)$, *then* (S) *holds with* $(u, \nabla u, T(\cdot,\nabla u))$.

Proof. $\eta = T(\cdot,\nabla u) \in L^\infty \subset H$; the equation extends to V by density, $F(\nabla u) < \infty$, and $\eta \in \partial F(\nabla u)$ by Lemmas 2.3 and 2.4(iii). □

Lemma 3.2 (Explicit flux). *Let* $u \in H_0^1(\Omega)$ *with* $F(\nabla u) < \infty$ *and* $\eta \in H$ *with* $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ *a.e. and* $(\eta, \nabla w) = (f, w)$ *for all* $w \in H_0^1(\Omega)$ (*e.g.* $\eta \in W^{1,\infty}$ *with* $-\operatorname{div}\eta = f$). *Then* (S) *holds.*

Proof. Lemma 2.3 and Proposition 2.2. □

Remark 3.3 (Radial test problems). On the unit disk with $f = c > 0$ and a radial Young integrand $G(x,t) = G_0(|x|,t)$ such that $g_0(s,\cdot)$ is continuous and strictly increasing on $(0,\infty)$ and $\inf_s g_0(s,t) \to \infty$ as $t \to \infty$, let $\tau(s) = 0$ if $cs/2 \le g_0(s,0)$ and otherwise $g_0(s,\tau(s)) = cs/2$. Then $u(x) = \int_{|x|}^1 \tau(s)\,ds$ is Lipschitz, $\eta(x) = -cx/2$ satisfies $\eta \in \partial\Phi(x,\cdot)(\nabla u)$ a.e. (Lemma 2.4(i) in the dead core of $e^t - 1$, (ii) elsewhere) and $-\operatorname{div}\eta = c$. By Lemma 3.2, (S) holds for these data, also for $e^t - 1 - t$, $e^t - 1$, variable exponents with $p^- < 2$ and discontinuous radial coefficients, which are outside the regularity theory below. The same holds for manufactured smooth solutions (Remark 2.16).

For the p-Laplacian (S) can be characterised. We use an abstract multiplier lemma, used again for Theorem C.

Lemma 3.4 (Multiplier lemma). *Let* X *be a real Hilbert space,* $R \subset X$ *a closed subspace, and* $\Psi\colon X \to \mathbb{R}$ *convex and bounded above on a ball centred at* $q_0 \in R$. *If* q_0 *minimises* Ψ *over* R, *there is* $\zeta \in \partial\Psi(q_0)$ *with* $\zeta \perp R$.

Proof. The directional derivative $\Psi'(q_0;z)$ is finite, sublinear, ≥ 0 on R and $\leq \Psi(q_0+z)-\Psi(q_0)$. By Hahn–Banach the zero functional on R extends to a linear $\zeta \leq \Psi'(q_0;\cdot)$ on X; boundedness of Ψ near q_0 makes ζ continuous, so $\zeta \perp R$ and $\zeta \in \partial\Psi(q_0)$. □

Proposition 3.5. *Let $G(t)=t^p/p$, $d\geq 2$ and $p\geq 2d/(d+2)$ (so that $W_0^{1,p}\hookrightarrow L^2$), and let u_p be the minimiser of J over $W_0^{1,p}(\Omega)$.*

(i) *If $1<p\leq 2$:* (S) *holds* $\iff$ *J attains its minimum on $H_0^1(\Omega)$* $\iff$ *$u_p\in H_0^1(\Omega)$.*
(ii) *If $p>2$ and $\nabla u_p\in L^{2p-2}$, then* (S) *holds. Conversely, on a Lipschitz domain,* (S) *implies $\nabla u_p\in L^{2p-2}$.*

Proof. (i) If J attains its minimum on V at u, apply Lemma 3.4 with $X=H$, $R=\nabla(V)$, $q_\ell\in R$ representing $\nabla w\mapsto(f,w)$, $\Psi(q)=F(q)-(q_\ell,q)$ (finite and bounded on balls, as $|q|^p\leq 1+|q|^2$) and $q_0=\nabla u$; $\eta=\zeta+q_\ell$ satisfies (2.1). Since J is continuous on $W_0^{1,p}$ and V is dense in it, a minimiser on V minimises on $W_0^{1,p}$, so $u=u_p$; the converse implications are clear. (ii) If $\nabla u_p\in L^{2p-2}$ the flux is in L^2 and the Euler–Lagrange equation extends to V; conversely a saddle point has $\nabla u^*\in L^p$, so $u^*\in W_0^{1,p}$ (Lipschitz domain), $u^*=u_p$ and $|\nabla u_p|^{p-1}\in L^2$. □

The argument of (i) works verbatim whenever F is finite on L^2 (e.g. $G(x,t)\leq A(x)+Ct^2$): then (S) holds if and only if J attains its minimum on H_0^1.

3.2. **A catalogue.** By Lemma 3.1, a gradient bound *up to the boundary* gives (S); interior regularity does not. Table 1 summarises what we found, together with the Lavrentiev question of Section 3.3.

Autonomous integrands with Lieberman's condition (includes t^p/p, $1<p<\infty$). Let $G=G(t)$ with $a(t)=g(t)/t\in C^1(0,\infty)$ and $-1<i_a\leq s_a<\infty$, where $i_a=\inf ta'/a$, $s_a=\sup ta'/a$; this is (L) with $\delta=1+i_a$, $g_0=1+s_a$, the structure condition of Lieberman [48], which implies Δ_2 and ∇_2. Cianchi and Maz'ya [24, Thm. 2.3] prove that on every bounded convex $\Omega\subset\mathbb{R}^d$, $d\geq 2$, and for every $f\in L^2(\Omega)$, the generalised solution has flux $a(|\nabla u|)\nabla u\in W^{1,2}(\Omega)$; [24, Thm. 2.4, Rem. 2.6] give the same on Lipschitz domains with $\partial\Omega\in W^{2,q}$, $q>1$ ($d=2$), or $\partial\Omega\in W^{2,d-1}$ ($d\geq 3$).

Proposition 3.6. *Under these hypotheses* (S) *holds if $d=2$, or if $d\geq 3$ and $i_a\geq -2/d$ (for t^p/p: $p\geq 2-2/d$).*

Proof. $\eta=a(|\nabla u|)\nabla u\in W^{1,2}\subset L^{2^*}$ (every L^s, $s<\infty$, if $d=2$). By (2.12), $g(t)\geq g(1)t^{1+i_a}$ for $t\geq 1$, so $|\nabla u|\leq 1+(|\eta|/g(1))^{1/(1+i_a)}\in L^2$ when $2^*(1+i_a)\geq 2$. Then $u\in H_0^1$, the generalised solution is the weak one [24, Rem. 2.8], the equation extends to H_0^1 since $\eta\in L^2$, $F(\nabla u)\leq\int|\nabla u||\eta|<\infty$, and $\eta\in\partial F(\nabla u)$. □

Globally bounded gradients, hence (S) by Lemma 3.1: [3, Cor. 1.4] ($C^{1,\alpha}$ domain, $f\in L^\sigma$, $\sigma>d$, $C^{1,\alpha}$ data, Hölder x-perturbations controlled by a *single* function b; a preprint), and Lieberman's boundary estimates [47, Thms. 1–2] (not available to us; its role is confirmed by [48, p. 320]), whose extension to condition (L) is asserted in [48, p. 320]; for $d\geq 3$, $f\in L^{d,1}$ and $\partial\Omega\in W^2L^{d-1,1}$, [22] (as recalled in [24, Rem. 2.6]; not checked in the original), and, for convex domains and p-type a with $2\leq p<d$, [23, Thm. 3.5], which also gives $|\nabla u|^{p-1}\in L^2$ already for $f\in L^{2d/(d+2)}$ [23, Thm. 4.1(ii)]. The examples in [23, §1] show that gradient bounds may fail on domains close to convex ones. Interior $C^{1,\alpha}$ regularity is classical [32, 56], and [48, Thm. 1.7] for (L); it does not give (S).

Variable exponents. For $p\in C^{0,\mu}(\overline{\Omega})$, $p^->1$ and $\partial\Omega\in C^{1,\beta}$, global $C^{1,\alpha}$ regularity of bounded weak solutions is due to Fan, see [39]; we could not check its hypotheses in the original, so (S) is conditional here. Interior results: [27, 1]. For log-Hölder or merely measurable exponents no gradient bound is known, and (S) must be assumed; when it holds, Corollary 2.15 gives strong convergence, without any continuity of p.

TABLE 1. (S) and the Lavrentiev gap. Column "(S)": known to hold (globally, unless stated); when it holds, Theorem B applies to all rows except the exponential one, where Theorem B(b) or Glowinski's theorem applies. Column "gap": whether $m_{\mathcal{H}} > m_{\mathcal{W}}$ is possible for some data. †: source not read by us (content taken from the cited secondary source).

Energy	(S)	Lavrentiev gap
t^p/p; autonomous G with (L)	yes: Ω convex, or $W^{2,q}$ if $d=2$, $f\in L^2$, $d=2$ (all p) or $i_a \geq -\frac{2}{d}$ [24, Thms. 2.3–2.4]; $C^{1,\alpha}$ domain, $f\in L^\sigma$, $\sigma > d$ [3, Cor. 1.4], [47]†. Fails if $u\notin H^1_0$ (Ex. 3.9)	no (zero data, star-shaped Ω; Prop. 3.11)
autonomous, no Δ_2 (e^t-1-t, e^t-1)	not known (explicit radial data only)	no (zero data, star-shaped Ω; Prop. 3.11)
$p(x)$, $p\in C^{0,\mu}(\overline{\Omega})$	yes, bounded u, $\partial\Omega\in C^{1,\beta}$ [39]†	no: log-Hölder [58]†, [31, §4.6]† (recalled in [7, p. 22])
$p(x)$ log-Hölder	not known	no [58]†, [31, §4.6]†
$p(x)$ continuous, not log-Hölder; discontinuous	not known	yes: [7, Thm. 30] (modulus $(\log(e+1/t))^{-\kappa}$, $\kappa<1$), [7, Thm. 29] (any $[p^-,p^+]$), checkerboard $p_1<2<p_2$, $d=2$ [57]† (recalled in [7, §1.1])
double phase, $a\in C^{0,\alpha}$, $q\leq p+\alpha\max\{1,\frac{p}{d}\}$	interior only ($f\equiv 0$) [25, 26, 9]; [29, Thm. 1.3]; must be assumed	no; global, Lipschitz Ω, data in $W^{1,q}$ [19, Thm. 2.3]
double phase, $q>p+\alpha\max\{1,\frac{p-1}{d-1}\}$	not known	yes [7, Thm. 32]; [38] ($p<d<d+\alpha<q$); see also [14]
two-phase, $a=\mathbf{1}_E$, E or $\Omega\setminus\overline{E}$ compactly inside, Lipschitz	not known (explicit radial data only)	no (Prop. 3.12(a))
two-phase, $a=\mathbf{1}_E$, checkerboard, $d=2$, $p<2<q$	not known	yes (Prop. 3.12(b))

Double phase and two-phase energies. For $a\in C^{0,\alpha}$ only *interior* gradient regularity is known, for $f\equiv 0$: $q/p < 1+\alpha/d$ [25, Thm. 1.2], $q/p\leq 1+\alpha/d$ [9, Thm. 1.1], bounded minimisers with $q\leq p+\alpha$ [26, 9]; with a right-hand side and $a\in W^{1,\sigma}$, $\sigma>d$, [29, Thm. 1.3]; general Musielak–Orlicz energies without right-hand side [46]. For discontinuous a nothing is available, except explicit solutions (Remark 3.3).
Exponential growth. For e^t-1-t ($s_a=\infty$) and e^t-1 ($g(0)=1$), [29, Thms. 1.4–1.5] require exponents >1 in the exponential, [24, 3] require $s_a<\infty$, and [59, p. 4] only remarks that their global method should extend to non-Δ_2 growth. No result known to us gives (S) for general data.
Failure of (S).. (S) fails whenever the minimiser is not in H^1_0 (Proposition 3.5(i)), even without any Lavrentiev phenomenon (Example 3.9).

3.3. **(S) and the Lavrentiev gap.** The literature on gradient regularity and on the Lavrentiev phenomenon overlaps (both fail for the same kind of non-autonomous integrands), and it is tempting to identify "(S) fails" with "there is a gap". They are different properties. In this subsection Ω is a bounded Lipschitz domain, $\ell(v)=\int_\Omega fv$ with $f\in L^d(\Omega)$ (so that ℓ is continuous on $W^{1,1}_0$), the boundary datum is zero (except in Proposition 3.12(b), where Lipschitz data ψ are used and the

infima are taken over $\psi+\mathcal{W}$, $\psi+H^1_0$, $\psi+C^\infty_c$), and

$$\mathcal{W}=\{v\in W^{1,1}_0(\Omega):F(\nabla v)<\infty\},\qquad m_{\mathcal{W}}=\inf_{\mathcal{W}}J,\qquad m_{H^1}=\inf_{H^1_0}J,\qquad m_{\mathcal{H}}=\inf_{C^\infty_c(\Omega)}J.$$

Then $m_{\mathcal{W}}\le m_{H^1}\le m_{\mathcal{H}}$; the *gap* is $m_{\mathcal{H}}-m_{\mathcal{W}}\ge 0$. If $\int_\Omega G(x,t)\,dx<\infty$ for every t, $m_{\mathcal{H}}$ is also the infimum over $W^{1,\infty}_0$ (truncate at levels $\pm\varepsilon$, mollify, dominated convergence). By Theorem A, under (S) the continuous iteration converges to a minimiser of J on H^1_0: the value it targets is m_{H^1}.

Proposition 3.7 (What (S) says about the infima)**.** *Assume* (S) *with* $(u^*,\nabla u^*,\eta^*)$*, and let* $D_0(\sigma)=-\int_\Omega\Phi^*(x,\sigma)\,dx$*.*

(i) $J(u^*)=m_{H^1}=\max\{D_0(\sigma):\sigma\in L^2(\Omega;\mathbb{R}^d),\ -\operatorname{div}\sigma=f$ *in* $\mathcal{D}'(\Omega)\}$*, attained at* η^**.*
(ii) *If* $G(x,t)\le A(x)+Ct^2$ *with* $A\in L^1$ *(subquadratic growth), then* $J(u^*)=m_{\mathcal{H}}$*.*
(iii) *If* $G(x,t)\ge ct^2-A(x)$ *with* $c>0$*,* $A\in L^1$ *(superquadratic growth), then* $\mathcal{W}\subset H^1_0$ *and* $J(u^*)=m_{\mathcal{W}}$*.*

Proof. (i) For admissible $\sigma\in L^2$ and $v\in H^1_0$ with $F(\nabla v)<\infty$, the Fenchel–Young inequality $\Phi(x,\nabla v)+\Phi^*(x,\sigma)\ge\sigma\cdot\nabla v$ and $\int\sigma\cdot\nabla v=\int fv$ give $J(v)\ge D_0(\sigma)$, with equality for (u^*,η^*) since $\eta^*\in\partial\Phi(x,\nabla u^*)$ a.e. (ii) F is finite, convex and l.s.c. on L^2, hence continuous; J is continuous on H^1_0, where C^∞_c is dense. (iii) If $v\in\mathcal{W}$ then $\nabla v\in L^2$; the zero extension of v lies in $W^{1,1}(\mathbb{R}^d)$ with gradient in L^2 and compact support, hence in $H^1(\mathbb{R}^d)$, and it vanishes outside Ω, which on a Lipschitz domain characterises $H^1_0(\Omega)$. □

Thus a dual certificate $\eta^*\in L^2$ certifies the minimum over H^1_0, not over $\mathcal{W}$: in the subquadratic regime (ii) the continuous method aims at $m_{\mathcal{H}}$, which may differ from $m_{\mathcal{W}}$; in the superquadratic regime (iii) it aims at $m_{\mathcal{W}}$, and a gap would make conforming discretisations inconsistent with the continuous method. Both situations are compatible with (S) as far as we know; the first one occurs:

Example 3.8 ((S) with a gap)**.** Let $\Omega=(-1,1)^d$, $1<p^-<p^+<2$, and let $p(\cdot)$ and the linear functional $S^\circ(w)=\int b^\circ\cdot\nabla w$ be those of [7, Thm. 29], so that $\inf_{\mathcal{W}}(F+S^\circ)<\inf_{C^\infty_c}(F+S^\circ)=0$. Since $b^\circ\in L^{p'(\cdot)}\subset L^2$ and $S^\circ=0$ on C^∞_c [7, Prop. 22], S° vanishes on H^1_0. As an element of H^{-1} the load is 0, so (S) holds with $(0,0,0)$ and $m_{H^1}=m_{\mathcal{H}}=0>m_{\mathcal{W}}$. The continuous iteration and P1 return 0; only a nonconforming method can see the gap. The example is degenerate (the load is not of the form $\int fv$); we know no example of (S) with a gap and $f\in L^d$.

Example 3.9 (No (S), no gap)**.** Let $p=\frac{3}{2}$, $\Omega=B_1\subset\mathbb{R}^2$, $f(x)=|x|^{-3/2}$ and $u(x)=4\log(1/|x|)$. Then $\eta=|\nabla u|^{-1/2}\nabla u=-2|x|^{-3/2}x$, $|\eta|=2|x|^{-1/2}\in L^3$, $-\operatorname{div}\eta=f$ in $\mathcal{D}'(B_1)$ (the flux through ∂B_ε is $-4\pi\varepsilon^{1/2}\to 0$), $u\in W^{1,3/2}_0(B_1)$ and $f\in L^{6/5}\subset(W^{1,3/2}_0)^*\cap H^{-1}$. So u minimises J on $W^{1,3/2}_0$, with $J(u)=(\frac{1}{p}-1)\int|\nabla u|^{3/2}=-\frac{32\pi}{3}$. Since $\nabla u\notin L^2$, (S) fails (Proposition 3.5(i)); since J is continuous on $W^{1,3/2}_0$, $m_{\mathcal{W}}=m_{\mathcal{H}}=J(u)$: there is no gap, and P1 and CR approximate the same minimiser. (Here $\eta\in L^2$; what fails is $u\in H^1_0$.)

What does hold are certificates in the other direction. Part (a) of the next proposition generalises the elementary observation that a Lipschitz $\mathcal{W}$-minimiser rules out a gap.

Proposition 3.10 (Flux certificates)**.** *Let* $u\in\mathcal{W}$ *and* $\eta\in L^\infty(\Omega;\mathbb{R}^d)$ *with* $\eta(x)\in\partial\Phi(x,\cdot)(\nabla u(x))$ *a.e. and* $\int\eta\cdot\nabla\varphi=\int f\varphi$ *for all* $\varphi\in C^\infty_c(\Omega)$*. Then* $J(u)=m_{\mathcal{W}}$*. If moreover* $u\in W^{1,\infty}$*, there is no gap. Consequently, if* (L) *holds,* $\operatorname{ess\,sup}_x g(x,1)<\infty$ *and* $m_{\mathcal{H}}>m_{\mathcal{W}}$*, the flux* $T(\cdot,\nabla u_{\mathcal{H}})$ *of the minimiser* $u_{\mathcal{H}}$ *of* J *on the closure of* C^∞_c *in* $W^{1,G(\cdot)}$ *is unbounded.*

Proof. Integrate $\Phi(x,\nabla v)\ge\Phi(x,\nabla u)+\eta\cdot\nabla(v-u)$ and use $\int\eta\cdot\nabla w=\int fw$ for $w\in W^{1,1}_0$ (density; both sides are $W^{1,1}$-continuous). For the last claim: under (L) the closure of C^∞_c is reflexive, J has a minimiser $u_{\mathcal{H}}$ there with $J(u_{\mathcal{H}})=m_{\mathcal{H}}$, and its Euler–Lagrange equation holds for $\varphi\in C^\infty_c$; a bounded flux would make $u_{\mathcal{H}}$ a $\mathcal{W}$-minimiser. □

When is there no gap? For autonomous energies on convex (or star-shaped) domains with zero data, never, and without any Δ_2 condition:

Proposition 3.11 (No gap for autonomous energies)**.** *Let $G = G(t)$ be any finite convex nondecreasing function with $G(0) = 0$, $f \in L^d$, zero Dirichlet data, and Ω bounded and star-shaped with respect to a ball (e.g. convex). For every $v \in \mathcal{W}$ there are $\varphi_k \in C_c^\infty(\Omega)$ with $\varphi_k \to v$ in $W^{1,1}$ and $F(\nabla\varphi_k) \to F(\nabla v)$; in particular $m_{\mathcal{W}} = m_{\mathcal{H}}$.*

Proof. Let the ball be $B_\rho(0)$; then $\operatorname{dist}(t\Omega, \partial\Omega) \geq (1-t)\rho$ for $t \in (0,1)$. Extend v by 0 and set $w_t(x) = t\, v(x/t)$: $\nabla w_t(x) = \nabla v(x/t)$, $\operatorname{supp} w_t \subset \overline{t\Omega}$ and $\int G(|\nabla w_t|) = t^d F(\nabla v)$. For $\varepsilon < (1-t)\rho$, $\varphi = \rho_\varepsilon * w_t \in C_c^\infty(\Omega)$ and, by Jensen's inequality, $\int G(|\nabla\varphi|) \leq \int \rho_\varepsilon * G(|\nabla w_t|) \leq F(\nabla v)$. As $t \uparrow 1$, $\varepsilon \downarrow 0$, $\varphi \to v$ in $W^{1,1}$, and Fatou's lemma gives $F(\nabla\varphi) \to F(\nabla v)$; ℓ is $W^{1,1}$-continuous. □

A localised version of this argument (partition of unity, inward translations, and a convex-combination trick replacing Δ_2) extends Proposition 3.11 to bounded Lipschitz domains; we only sketch it and do not use it. Two-phase energies with a discontinuous coefficient behave differently depending on the geometry:

Proposition 3.12 (Two-phase energies)**.** *Let $\Phi(x,\xi) = \frac{1}{p}|\xi|^p + \mathbf{1}_E(x)\frac{1}{q}|\xi|^q$, $1 < p < q$, $f \in L^d$.*

(a) *If $E \Subset \Omega$, or $\Omega \setminus \overline{E} \Subset \Omega$, with E, $\Omega \setminus \overline{E}$ and Ω Lipschitz (e.g. $E = \{\frac{1}{2} < |x| < 1\}$ in the unit disk), then $m_{\mathcal{W}} = m_{\mathcal{H}}$.*

(b) *If $d = 2$, $\Omega = (-1,1)^2$, $E = \{x_1 x_2 > 0\}$ and $p < 2 < q$, there are Lipschitz boundary data (and $f = 0$) with $m_{\mathcal{W}} < m_{\mathcal{H}}$.*

Proof. (a) Let $v \in \mathcal{W}$, so $v|_E \in W^{1,q}(E)$. Extend $v|_E$ with a Stein extension operator bounded simultaneously in $W^{1,p}$ and $W^{1,q}$ (if $\Omega \setminus \overline{E} \Subset \Omega$, extend from $\mathbb{R}^d \setminus \overline{(\Omega \setminus \overline{E})}$ the function equal to v on E and 0 outside Ω; if $E \Subset \Omega$, extend and cut off), obtaining $w \in W_0^{1,q}(\Omega)$ with $w = v$ on E. Then $z = v - w \in W_0^{1,p}(\Omega \setminus \overline{E})$. Approximating w in $W^{1,q}$ by $C_c^\infty(\Omega)$ and z in $W^{1,p}$ by $C_c^\infty(\Omega \setminus \overline{E})$, the sum converges in energy. (b) Zhikov's function u (0-homogeneous, ∇u supported in $\{x_1 x_2 < 0\}$, $|\nabla u| \sim |x|^{-1}$) and separating field b (supported in E, $|b| \sim |x|^{-1}$, $b \cdot \nabla u = 0$, $\operatorname{div} b = 0$ in $\mathcal{D}'$), as recalled in [7, §1.1], satisfy $u \in W^{1,\Phi}$ (since $p < 2$) and $b \in L^{\Phi^*}$ (since $\Phi^*(x,s) \leq s^{q'}/q'$ on E and $q' < 2$). For large t and $s = t^\theta$, $p - 1 < \theta < q - 1$, $F(tu) + F^*(sb) \leq C(t^p + t^{\theta q'}) < ts$, which is condition (10) of [7]; [7, Thm. 28], applied to Zhikov's pair (whose properties we take from [7, §1.1] and [57]; not checked in the original), then gives boundary data $g = t\, u^\partial$ (the boundary values of tu) for which the $\mathcal{W}$- and $\mathcal{H}$-minimisers differ and their energies satisfy $F(h_{\mathcal{W}}) < F(h_{\mathcal{H}})$. □

The proof carries over to the configuration rotated by $\pi/4$, $E = \{|x_1| < |x_2|\}$, used in Section 6; but a gap for the specific data λu_0 used there is *not proved*: it is inferred numerically ([8] and Section 6.5).

In the planar checkerboards of Proposition 3.12(b) and of Zhikov's $p(x)$ (with the exponent > 2 on E), there is moreover an obstruction to H^1: an $H^1(B_\rho)$ function whose restrictions to two closed sectors with vertex 0 have continuous representatives has the same limit at 0 from both sectors (since $\int |\nabla v|^2 \geq \int_0^\rho \frac{1}{r} \int_0^{2\pi} |\partial_\theta v|^2 \, d\theta \, dr$ and $\int_0 \frac{dr}{r} = \infty$). Functions of $H^1 \cap \mathcal{W}$ are continuous at 0 on the two components of $\overline{E}$ (Morrey, since the exponent there is > 2), with equal limits, whereas the singular competitors of [57, 7] jump. If, as in Zhikov's analysis, the closure of smooth functions consists exactly of the functions of $\mathcal{W}$ without such a jump (we have not checked this in the original), then $m_{H^1} = m_{\mathcal{H}}$ there: under (S) the continuous method aims at the $\mathcal{H}$-minimiser, like P1, and the $\mathcal{W}$-minimiser is not in H^1.

Remark 3.13 (Summary). (1) Convergence of the continuous iteration ⇔ (S) ⇔ bounded iterates (Theorem A, Proposition 2.11); (S) concerns H_0^1 and can fail without any gap (Example 3.9). (2) The gap concerns $\mathcal{H}$ versus $\mathcal{W}$; (S) neither implies nor excludes it in general (Proposition 3.7,

Example 3.8). (3) A gap forces an unbounded flux of the $\mathcal{H}$-minimiser (Proposition 3.10); its discrete counterpart, a computable diagnostic, is Theorem 5.7. (4) Whatever the status of (S), when there is a gap the relevant question becomes question (II) of the introduction: which discrete problem approximates $m_{\mathcal{W}}$? This requires first a convergence theory for the discrete iteration that does not depend on regularity.

4. The discrete problems: convergence without regularity

4.1. **Two discretisations with the same flux space.** Let $\Omega \subset \mathbb{R}^2$ be a bounded polygon and $\mathcal{T}_h$ a conforming triangulation of Ω (for the disk in Section 6, Ω is replaced by the inscribed polygon Ω_h; everything in this section concerns the triangulated polygon). $\mathcal{E}_h^\circ$ and $\mathcal{E}_h^\partial$ denote the interior and boundary edges, m_E the midpoint of E, x_T the barycentre of T. The flux space is

$$H_h = \{q\colon \Omega \to \mathbb{R}^2 :\ q|_T \text{ constant } \forall T \in \mathcal{T}_h\}, \qquad (q,q')_h = \sum_T |T|\, q_T \cdot q'_T,$$

and for u we use either the conforming space or the Crouzeix–Raviart space [28]:

$$\begin{aligned} \mathrm{P1}_h &= \{v \in C(\overline{\Omega}) :\ v|_T \in \mathbb{P}_1\ \forall T\}, \qquad V_h^{\mathrm{P1}} = \{v \in \mathrm{P1}_h :\ v = 0 \text{ on } \partial\Omega\},\\ \mathrm{CR}_h &= \{v\colon v|_T \in \mathbb{P}_1\ \forall T,\ v \text{ continuous at } m_E\ \forall E \in \mathcal{E}_h^\circ\},\\ V_h^{\mathrm{CR}} &= \{v \in \mathrm{CR}_h :\ v(m_E) = 0\ \forall E \in \mathcal{E}_h^\partial\}. \end{aligned}$$

For $v \in \mathrm{CR}_h$, $\nabla_h v \in H_h$ is the elementwise gradient (the gradient on $\mathrm{P1}_h$), and $B_h = \nabla_h\colon V_h \to H_h$. We assume that $G(x,\cdot)$ is a Young function for *every* x, write $G_T = G(x_T,\cdot)$, $\Phi_T(\xi) = G_T(|\xi|)$, and set

$$F_h(q) = \sum_T |T|\, G_T(|q_T|), \qquad J_h(v) = F_h(\nabla_h v) - \ell_h(v),$$

where ℓ_h is a linear functional on V_h (in the code, the barycentre rule for P1 and the edge-midpoint rule for CR; both equal $\int_\Omega fv$ when f is constant, and, for piecewise constant f, when f is evaluated inside each triangle). When the mesh *resolves* G, i.e.

$$G(x,\cdot) = G_T \quad \text{for a.e. } x \in T, \text{ for every } T \in \mathcal{T}_h, \tag{4.1}$$

$F_h(\nabla_h v) = \int_\Omega G(x, |\nabla_h v|)\,dx$ is the exact energy of the broken gradient.

The algorithm (2.3)–(2.5) is applied to $(V_h, H_h, B_h, F_h, \ell_h)$. The flux space, the pointwise step (one scalar equation (2.8) per triangle, Lemma 2.6) and the multiplier step are *identical* for P1 and CR; only the linear step changes: $rA\,U_n = b_h + [(r\nu_{n-1} - \eta_n, \nabla_h\phi_i)_h]_i$, where ϕ_i runs over the nodal basis of V_h^{P1} (unknowns: interior vertices) or over the CR basis ϕ_E, $E \in \mathcal{E}_h^\circ$, $\phi_E(m_{E'}) = \delta_{EE'}$ (unknowns: interior edges), and A is the stiffness matrix, factorised once.

Lemma 4.1 (∇_h is injective)**.** *$\|\nabla_h v\|_h$ is a norm on V_h^{P1} and on V_h^{CR}; hence A is symmetric positive definite and $(V_h, H_h, B_h, F_h, \ell_h)$ satisfies (A1)–(A3) with $\|v\|_{V_h} := \|\nabla_h v\|_h$, F_h finite, convex and continuous.*

Proof. Let $\nabla_h v = 0$; then v is constant on each triangle. In the CR case two triangles sharing an edge $E \in \mathcal{E}_h^\circ$ take the same value at m_E, hence the same constant. Any two triangles are joined by a chain of triangles in which consecutive ones share an edge (a path in the connected polygon avoiding the finitely many vertices passes from triangle to triangle through interiors of edges), so v is one constant, and it vanishes at the midpoint of a boundary edge. The P1 case is classical. F_h is a finite sum of finite convex functions of the blocks q_T. □

An h-uniform version of Lemma 4.1 (a broken Poincaré–Friedrichs inequality) is not needed for Theorem C; it is needed for the limit $h \to 0$, see [53, (4.4)], [17] and [8, Lemma 3].

Nonhomogeneous data. For a boundary datum ψ the discrete admissible sets are

$$\mathrm{P1}_{h,\psi} = \{v \in \mathrm{P1}_h : \ v(z) = \psi(z) \text{ at the boundary vertices}\},$$

$$\mathrm{CR}_{h,\psi} = \Big\{v \in \mathrm{CR}_h : \ v(m_E) = \bar{\psi}_E := \frac{1}{|E|}\int_E \psi\, ds \ \ \forall E \in \mathcal{E}_h^\partial\Big\}.$$

With a lifting $\tilde{u}_h$ (prescribed boundary values, zero interior unknowns) one writes $u = \tilde{u}_h + v$, $v \in V_h$, and applies everything to $\tilde{F}_h(q) = F_h(q + \nabla_h \tilde{u}_h)$, which is finite, convex, nonnegative and superlinear when F_h is; (A2) only asks F to be proper, and any initialisation is admissible (Remark 2.10).

4.2. **Theorem C.**

Theorem C. *Let $V_h = V_h^{\mathrm{P1}}$ or $V_h = V_h^{\mathrm{CR}}$ (possibly with nonhomogeneous data as above), let $G(x,\cdot)$ be a Young function for every x, and assume*

$$\lim_{t\to\infty} G_T(t)/t = +\infty \qquad \text{for every } T \in \mathcal{T}_h. \tag{4.2}$$

Then:

- (i) *J_h has a minimiser u_h in the discrete admissible set; it is unique if every G_T is strictly convex.*
- (ii) *There is $\eta_h \in H_h$ such that $(u_h, \nabla_h u_h, \eta_h)$ is a saddle point of the discrete augmented Lagrangian for every $r > 0$; equivalently, $\eta_{h,T} \in \partial\Phi_T(\nabla_h u_h|_T)$ for every T and $(\eta_h, \nabla_h w)_h = \ell_h(w)$ for every $w \in V_h$.*
- (iii) *For every $r > 0$ and every initialisation $\nu_0, \eta_1 \in H_h$, the iterates converge: $u_n \to \bar{u}_h$, $\nu_n \to \nabla_h \bar{u}_h$, $\eta_n \to \bar{\eta}_h$, where $(\bar{u}_h, \nabla_h \bar{u}_h, \bar{\eta}_h)$ is a discrete saddle point and $\bar{u}_h$ minimises J_h; $\nabla_h u_n - \nu_n \to 0$, and the identity (2.9) holds with $(\cdot,\cdot)_h$, so that E_n is nonincreasing.*

Proof. By Lemma 4.1 and the lifting it suffices to treat V_h with $\tilde{F}_h$, which we still call F_h.

(i) Let $L = \sup\{|\ell_h(v)| : \|\nabla_h v\|_h \le 1\}$ ($< \infty$, V_h being finite-dimensional) and $a_{\min} = \min_T |T|$. For $q \in H_h$, $\|q\|_h^2 \le \max_T |q_T| \sum_T |T||q_T| \le a_{\min}^{-1}(\sum_T |T||q_T|)^2$. Given $M > 0$, (4.2) gives t_M with $G_T(t) \ge Mt - Mt_M$ for all $t \ge 0$ and all (finitely many) T. Hence $J_h(v) \ge (Ma_{\min}^{1/2} - L)\|\nabla_h v\|_h - Mt_M|\Omega| - C_\psi$, where C_ψ accounts for the lifting (G_T is nondecreasing and $|q_T + \nabla_h \tilde{u}_h|_T| \ge |q_T| - |\nabla_h \tilde{u}_h|_T|$). For $Ma_{\min}^{1/2} > L$, J_h is coercive and continuous on the finite-dimensional V_h, so it attains its minimum. If every G_T is strictly convex, it is strictly increasing, so Φ_T is strictly convex (for $|\xi_1| = |\xi_2|$, $\xi_1 \ne \xi_2$, use $|(\xi_1+\xi_2)/2| < |\xi_1|$); hence F_h is strictly convex and so is $J_h = F_h \circ \nabla_h - \ell_h$, ∇_h being injective.

(ii) Let $R = \nabla_h(V_h)$ and $q_\ell \in R$ with $(q_\ell, \nabla_h w)_h = \ell_h(w)$. $\Psi(q) = F_h(q) - (q_\ell, q)_h$ is finite and convex on H_h, hence bounded above near every point, and $\nabla_h u_h$ minimises Ψ on R. Lemma 3.4 gives $\zeta \in \partial\Psi(\nabla_h u_h)$, $\zeta \perp R$; $\eta_h = \zeta + q_\ell$ satisfies $\eta_h \in \partial F_h(\nabla_h u_h)$ and $(\eta_h, \nabla_h w)_h = \ell_h(w)$. Since F_h is a sum of functions of the blocks q_T with weights $|T| > 0$, $\eta_h \in \partial F_h(q)$ iff $\eta_{h,T} \in \partial\Phi_T(q_T)$ for all T. By Proposition 2.2 (which uses only (A1)–(A3)) this is a saddle point for every $r > 0$. This is the finite-dimensional duality of [43, Ch. VI, §4.2].

(iii) By (ii), (S) holds for the discrete data, and Theorem A, whose proof uses only (A1)–(A3), gives weak, hence (finite dimension) strong, convergence to a saddle point, for any initialisation (Remark 2.10). By Proposition 2.2, $\bar{u}_h$ minimises J_h. □

Remark 4.2 (What changes from P1 to CR; what is not claimed). Only two points depend on the space: the injectivity of ∇_h (Lemma 4.1, where continuity at the midpoints of interior edges and vanishing at the midpoints of boundary edges replace the continuity of P1 functions), and the representation of the boundary datum (edge means instead of nodal values). No discrete inf–sup condition and no h-uniform estimate is involved. Theorem C(iii) also follows from [36, Thm. 8]; Glowinski's finite-dimensional [43, Ch. VI, Thm. 5.2] requires a strictly convex C^1 part.

Superlinearity cannot be dropped (for $G(t) = t$ and large ℓ_h, J_h is unbounded below). Theorem C says nothing about the number of iterations or its dependence on h and r (Section 6), nor about the limit $h \to 0$, which is the subject of Section 5.

Remark 4.3 (An exactly equilibrated multiplier). By (2.3), the field

$$\tilde{\eta}_n := \eta_n + r(\nabla_h u_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1}) \in H_h \tag{4.3}$$

satisfies $(\tilde{\eta}_n, \nabla_h w)_h = \ell_h(w)$ for all $w \in V_h$ *at every iteration* (up to the round-off of the linear solve), whereas η_n itself is equilibrated only in the limit. By Theorem C(iii), $\tilde{\eta}_n \to \bar{\eta}_h$. This is used in Section 5.3.

5. Consistency and the Lavrentiev phenomenon: a guaranteed bracket

Theorem C settles question (I) at the discrete level. Question (II) is about the limit $h \to 0$ of the discrete minima, and here the Lavrentiev gap, not (S), is decisive. Throughout this section $\Omega \subset \mathbb{R}^2$ is a polygon, $\psi \in W^{1,1}(\Omega)$ the boundary datum, $f \in L^\infty(\Omega)$,

$$\mathcal{W}_\psi = \{v \in W^{1,1}(\Omega) : v - \psi \in W^{1,1}_0(\Omega)\}, \qquad \mathcal{H}_\psi = \mathcal{W}_\psi \cap W^{1,\infty}(\Omega),$$

$J(v) = \int_\Omega G(x, |\nabla v|)\,dx - \int_\Omega fv\,dx$ for $v \in \mathcal{W}_\psi$ (possibly $+\infty$), $m_{\mathcal{W}} = \inf_{\mathcal{W}_\psi} J$ and $m_{\mathcal{H}} = \inf_{\mathcal{H}_\psi} J$. This is the space $H = W \cap W^{1,\infty}$ of [8]; for $\psi = 0$ and G bounded on bounded sets it gives the same $m_{\mathcal{H}}$ as the smooth functions of Section 3.3 (Remark 5.2).

5.1. **Without a gap.** For autonomous continuous superlinear integrands on polygons, if there is no gap (which is always the case for scalar autonomous convex problems with zero data on convex domains, Proposition 3.11), a subsequence of P1 minimisers converges weakly in $W^{1,1}$ to a $\mathcal{W}$-minimiser and the energies converge [53, Thm. 3.1] ($f \in L^\infty$, $W^{1,\infty}$ boundary data, uniformly shape-regular meshes); CR minimisers with edge-mean data converge in the same sense without any assumption on the gap [53, Thm. 4.1] (same hypotheses), with strong convergence for strictly convex integrands. For integrands continuous in x and homogeneous data, conforming energies converge to the infimum over Lipschitz functions [21, Thm. 2.1]. Rates are known only under regularity of the solution: for the p-Laplacian [10, 50, 35], for $\Delta_2 \cap \nabla_2$ N-functions in the quasi-norm $\|\mathcal{V}(\nabla u) - \mathcal{V}(\nabla u_h)\|$ when $\mathcal{V}(\nabla u) \in W^{1,2}$ [34, 33], for Hölder and log-Hölder $p(\cdot)$ [16, 30], and for CR with Hölder $p(\cdot)$ [6]. None of these applies to discontinuous coefficients or to $e^t - 1 - t$. For double phase energies with $a \in C^{0,\alpha}$ in the no-gap range of [19, Thm. 2.3] and boundary data in $W^{1,q}$, Proposition 5.1 below with a majorant sampling of a (Remark 5.2(c)) gives convergence of the P1 minima to $m_{\mathcal{W}}$; with barycentre sampling we have no proof.

5.2. **The conforming energy is an upper bound.**

Proposition 5.1 (P1: upper bound and limit)**.** *Assume (4.1), that $\psi|_{\partial\Omega}$ is affine on every boundary edge of $\mathcal{T}_h$, and that $\ell_h(v) = \int_\Omega fv$ for $v \in \mathrm{P1}_h$ (e.g. f piecewise constant and the barycentre rule, or $f = 0$).*

- (i) *For every $v_h \in \mathrm{P1}_{h,\psi}$, $v_h \in \mathcal{H}_\psi$ and $J_h(v_h) = J(v_h) \geq m_{\mathcal{H}} \geq m_{\mathcal{W}}$. In particular this holds for every iterate u_n of the P1 iteration.*
- (ii) *Let $(\mathcal{T}_h)$ be a shape-regular family of triangulations satisfying these assumptions for every h (e.g. the criss-cross meshes of Section 6, which resolve $|x_1| = |x_2|$), with maximal diameter $h \to 0$; assume $\sup_x G(x, M) < \infty$ for every $M > 0$ and $\mathcal{H}_\psi \neq \emptyset$. Then $\min_{\mathrm{P1}_{h,\psi}} J_h \to m_{\mathcal{H}}$.*

Proof. (i) v_h is Lipschitz; on each boundary edge v_h and ψ are affine with equal endpoint values, so $v_h - \psi \in W^{1,1}_0(\Omega)$. By (4.1), $F_h(\nabla v_h) = \int_\Omega G(x, |\nabla v_h|)$. (ii) By (i), $\liminf_h \min J_h \geq m_{\mathcal{H}}$. Let $v \in \mathcal{H}_\psi$ with $J(v) < \infty$ and $I_h v$ its nodal interpolant; $I_h v \in \mathrm{P1}_{h,\psi}$. By [21, Lemma 2.1], $\|\nabla I_h v\|_{L^\infty} \leq C\|v\|_{W^{1,\infty}}$ with C depending only on the shape regularity; $I_h v \to v$ uniformly and $\nabla I_h v \to \nabla v$ a.e. Dominated convergence (majorant $\sup_x G(x, C\|v\|_{W^{1,\infty}})$) gives $J(I_h v) \to J(v)$, so $\limsup_h \min J_h \leq J(v)$. □

Remark 5.2. (a) Under the assumptions of Proposition 5.1(ii) with $\psi \in W^{1,\infty}$, $m_{\mathcal{H}}$ equals the infimum over $\psi + C_c^\infty(\Omega)$: for $w = v - \psi$ Lipschitz with zero trace, $w_\varepsilon = (w-\varepsilon)_+ - (w+\varepsilon)_-$ has compact support and $|\nabla w_\varepsilon| \le |\nabla w|$; mollify and use dominated convergence. (b) [21, Thm. 2.1] assumes an integrand continuous on $\overline{\Omega} \times \mathbb{R}^m \times \mathbb{R}^{m\times n}$ and homogeneous data, so it does not cover discontinuous $p(\cdot)$ or $a(\cdot)$ literally; this is why Proposition 5.1 is stated (its proof is that of [21]). (c) If (4.1) fails, (i) still holds for a *majorant* sampling, $G_T \ge G(x,\cdot)$ on T (e.g. $x_T \in \operatorname{argmax}_T a$ for $t^p/p + a(x)t^q/q$): then $J_h(v_h) \ge J(v_h) \ge m_{\mathcal{H}}$. If moreover $\sup_{x\in T}(G_T(t) - G(x,t)) \le \omega(h)\Gamma(t)$ with $\omega(h) \to 0$ and Γ bounded on bounded sets (e.g. a uniformly continuous), the proof of (ii) gives again $\min J_h \to m_{\mathcal{H}}$.

Thus with a gap the conforming method converges, but to the wrong value $m_{\mathcal{H}} > m_{\mathcal{W}}$; this is the failure of conforming finite elements described by Carstensen and Ortner [21]. Since the P1 iteration converges for every r (Theorem C), the failure is entirely one of consistency, not of the solver. Balci, Ortner and Storn [8] showed that Crouzeix–Raviart elements avoid it:

Remark 5.3 (Convergence of CR energies to $m_{\mathcal{W}}$). [8, Thm. 1] assumes a Carathéodory integrand $\phi(x,\xi)$, convex in ξ, with $-c_0 + c_1|\xi|^{p_-} \le \phi(x,\xi) \le c_2|\xi|^{p_+} + c_0$, $1 < p_- < p_+ < \infty$, a polygon, $\psi \in W^{1,p_+}(\Omega)$, $f \in L^{p_-'}$, CR boundary values equal to edge means, the one-point quadrature $\phi(x_T,\cdot)$ on T, and either their conditions (A1)–(A3) or (B); it concludes that the discrete CR energies converge to $\min_{\mathcal{W}_\psi} J$, with convergence of the minimisers (strong for strictly convex integrands). For the checkerboard $p(\cdot)$ with $(p_-,p_+) = (3/2,3)$ and for the two-phase integrand $t^{3/2}/(3/2) + \mathbf{1}_{\{|x_1|<|x_2|\}}t^3/3$ on meshes satisfying (4.1), $\phi_h = \phi$ a.e., so (B) and (A1) with $c = 1$ hold trivially, (A2) holds with $C = 8$, and the growth condition holds with $c_1 = 1/3$, $c_2 = c_0 = 1$; the datum λu_0 of Section 6.4 has the trace of $\psi = \lambda x_2 \in W^{1,\infty}$. These are Experiments 1 and 3 of [8], where the authors state that the assumptions hold. Integrands of exponential growth violate the upper bound, so [8, Thm. 1] does not apply to them; for *autonomous* convex superlinear integrands (including $e^t - 1 - t$), CR convergence is [53, Thm. 4.1] ($f \in L^\infty$, $W^{1,\infty}$ boundary data, uniformly shape-regular meshes).

Combined with Theorem C, Remark 5.3 gives Glowinski's method with CR elements a convergence theory in the Lavrentiev regime *under the assumptions of* [8, Thm. 1] (polynomial two-sided growth, resolved coefficients or minimum quadrature): for every mesh the iteration converges for every r to the CR minimiser, and the CR minima converge to $m_{\mathcal{W}}$ as $h \to 0$ (no rates, and nothing on joint limits in the number of iterations and h; exponential integrands are excluded). Two practical questions remain: how can one tell from computations whether there is a gap, and how far is a computed energy from $m_{\mathcal{W}}$? A lower CR energy alone answers neither, since it is lower also without a gap (Remark 5.6(e)). The next theorem gives a quantitative answer.

5.3. **The Crouzeix–Raviart energy and a guaranteed lower bound.** If $\phi(\xi) = \gamma(|\xi|)$ with γ a Young function, then $\phi^*(\zeta) = \gamma^*(|\zeta|)$, and the Fenchel–Young inequality $\gamma(|\xi|) + \gamma^*(|\zeta|) \ge \zeta\cdot\xi$ holds with equality if and only if $\zeta \in \partial\phi(\xi)$ [37]. If γ is superlinear, γ^* is finite, convex and continuous. Let

$$\mathrm{RT}_0(\mathcal{T}_h) = \{\sigma \in H(\mathrm{div},\Omega) :\ \sigma|_T(x) = a_T + b_T\, x,\ a_T \in \mathbb{R}^2,\ b_T \in \mathbb{R}\ \ \forall T\}$$

be the lowest-order Raviart–Thomas space [54]; for $\sigma \in \mathrm{RT}_0(\mathcal{T}_h)$ the normal component $(\sigma\cdot n)_E$ is constant on each edge.

Lemma 5.4 (Generalised Marini identity)**.** *Let f_h be piecewise constant on $\mathcal{T}_h$ and let $\zeta \in H_h$ satisfy*

$$(\zeta, \nabla_h w)_h = \int_\Omega f_h w\, dx \qquad \forall w \in V_h^{\mathrm{CR}}. \tag{5.1}$$

Then $\sigma := \zeta - \frac{f_h}{2}(x - x_T)$ on each T belongs to $\mathrm{RT}_0(\mathcal{T}_h)$ and $\operatorname{div}\sigma = -f_h$. If $f_h = 0$, $\sigma = \zeta$.

Proof. On T, $\sigma = a_T + b_T x$, so $\sigma \cdot n$ is constant on each edge of T and $\operatorname{div}\sigma|_T = -f_T$. Fix $E \in \mathcal{E}_h^\circ$, $E = \partial T_+ \cap \partial T_-$, and ϕ_E. On $T \ni E$, ϕ_E is affine, $\int_{E'} \phi_E = |E'|\delta_{EE'}$ for the edges E' of T, $\int_T \phi_E = |T|/3$ and $\int_T (x - x_T) = 0$. Green's formula on T gives $|E|(\sigma|_T \cdot n_T)|_E = \int_T \sigma \cdot \nabla\phi_E + \int_T \phi_E \operatorname{div}\sigma = |T|\zeta_T \cdot \nabla\phi_E|_T - f_T \int_T \phi_E$. Summing over $T_\pm$ and using (5.1) with $w = \phi_E$, the jump of $\sigma \cdot n$ across E vanishes; a piecewise polynomial field with continuous normal components belongs to $H(\operatorname{div}, \Omega)$. □

For homogeneous Dirichlet data and the exact discrete flux of a $p(\cdot)$-problem, Lemma 5.4 and the identity of Theorem D(ii) below are the generalised Marini formula and the discrete strong duality of [6, Prop. 2.37] (whose discrete dual uses $G_T^*(|\Pi_T\sigma|)$; the two coincide when $f = 0$); see [11] for general convex problems and [51, 5] for the linear case; guaranteed lower energy bounds from CR elements for problems without a gap are studied in [20] (not read by us). What is added here, as far as we know, is the comparison with the *continuous* infimum $m_{\mathcal{W}}$, which holds also in the presence of a Lavrentiev gap, the boundary term for edge-mean data, the certificate at every iterate of the ADMM, and the bracket with the P1 energy.

Theorem D (Guaranteed lower bound)**.** *Let Ω be a polygon triangulated by $\mathcal{T}_h$ with (4.1), $\psi \in W^{1,1}(\Omega)$, $f \in L^\infty(\Omega)$. For $\sigma \in \mathrm{RT}_0(\mathcal{T}_h)$ with $\operatorname{div}\sigma = -f$ define*

$$D(\sigma) = \sum_{E \in \mathcal{E}_h^\partial} (\sigma \cdot n)_E \int_E \psi \, ds - \sum_{T \in \mathcal{T}_h} \int_T G_T^*(|\sigma(x)|) \, dx \ \in [-\infty, \infty) \tag{5.2}$$

(such σ exist only if f is piecewise constant on $\mathcal{T}_h$).

(i) (Weak duality.) *$J(v) \geq D(\sigma)$ for every $v \in \mathcal{W}_\psi$; hence $m_{\mathcal{W}} \geq D(\sigma)$. If moreover f is piecewise constant and $\ell_h(v) = \int fv$ on CR_h, then $J_h(v_h) \geq D(\sigma)$ for every $v_h \in \mathrm{CR}_{h,\psi}$.*

(ii) (The CR minimum.) *Let $f = f_h$ be piecewise constant, $\ell_h(v) = \int_\Omega f_h v$, every G_T superlinear, u_h a minimiser of J_h on $\mathrm{CR}_{h,\psi}$, η_h as in Theorem C(ii), and σ_h the field of Lemma 5.4 with $\zeta = \eta_h$. Then*

$$J_h(u_h) - D(\sigma_h) = \sum_T \Big[\int_T G_T^*(|\sigma_h|) \, dx - |T| \, G_T^*(|\eta_{h,T}|) \Big] \ \geq 0.$$

In particular, if $f = 0$ then $\sigma_h = \eta_h$ and

$$\min_{\mathrm{CR}_{h,\psi}} F_h(\nabla_h \cdot) = F_h(\nabla_h u_h) = D(\eta_h) \leq m_{\mathcal{W}}.$$

(iii) (A certificate at every iteration.) *Under the assumptions of (ii), let (u_n, ν_n, η_n) be the CR iterates and σ_n the field of Lemma 5.4 with $\zeta = \tilde{\eta}_n$ from (4.3). Then $D(\sigma_n) \leq m_{\mathcal{W}}$ and $D(\sigma_n) \leq J_h(u_n)$ for every n, and $D(\sigma_n) \to D(\bar{\sigma}_h)$, the value in (ii) for the limit saddle point; if $f = 0$, $D(\sigma_n) \to \min_{\mathrm{CR}_{h,\psi}} F_h(\nabla_h \cdot)$.*

No growth condition beyond (4.2) (used only for existence in (ii)–(iii)), no Δ_2 condition and no absence of a gap are assumed.

Proof. (i) Let $v \in \mathcal{W}_\psi$ with $J(v) < \infty$ and $\int G_T^*(|\sigma|) < \infty$. Pointwise, $G(x, |\nabla v|) \geq \sigma \cdot \nabla v - G_T^*(|\sigma|)$ a.e. in T, by (4.1) and Fenchel–Young; all terms are integrable. Since $\sigma, \operatorname{div}\sigma \in L^\infty$, $\int_\Omega \sigma \cdot \nabla w = \int_\Omega f w$ for $w \in C_c^\infty(\Omega)$, hence for $w \in W_0^{1,1}(\Omega)$; with $w = v - \psi$,

$$J(v) \geq \int_\Omega \big(\sigma \cdot \nabla\psi + \psi \operatorname{div}\sigma\big) \, dx - \int_\Omega G^*(x, |\sigma|) \, dx.$$

Green's formula on each triangle, the equality of the traces of ψ from both sides of interior edges and the continuity of $\sigma \cdot n$ give $\int_\Omega (\sigma \cdot \nabla\psi + \psi \operatorname{div}\sigma) = \sum_{E \in \mathcal{E}_h^\partial} (\sigma \cdot n)_E \int_E \psi$. For $v_h \in \mathrm{CR}_{h,\psi}$: by

Fenchel–Young with $\Pi_T\sigma := |T|^{-1}\int_T \sigma$ and Jensen,

$$J_h(v_h) \geq \sum_T |T|\big(\Pi_T\sigma\cdot\nabla v_h|_T - G_T^*(|\Pi_T\sigma|)\big) - \int f v_h \geq \sum_T \int_T \sigma\cdot\nabla v_h - \int f v_h - \sum_T \int_T G_T^*(|\sigma|),$$

and elementwise Green's formula turns $\sum_T \int_T \sigma\cdot\nabla v_h - \int f v_h$ into $\sum_T \int_{\partial T} v_h\,\sigma\cdot n_T$. On an interior edge $\sigma\cdot n$ is constant and continuous and $\int_E v_h|_{T_\pm} = |E|\,v_h(m_E)$ from both sides, so the contributions cancel; on a boundary edge $\int_E v_h = |E|\,\bar\psi_E = \int_E \psi$.

(ii) By Theorem C(ii), η_h satisfies (5.1), so $\sigma_h \in \mathrm{RT}_0(\mathcal{T}_h)$, $\operatorname{div}\sigma_h = -f_h$ and $\Pi_T\sigma_h = \eta_{h,T}$. The computation in (i) with $v_h = u_h$, now with equality in Fenchel–Young ($\eta_{h,T} \in \partial\Phi_T(\nabla u_h|_T)$), gives $J_h(u_h) = \sum_{E\in\mathcal{E}_h^\partial}(\sigma_h\cdot n)_E \int_E \psi - \sum_T |T| G_T^*(|\eta_{h,T}|)$, which is the claim; the bracket is ≥ 0 by Jensen, and vanishes if $f_h = 0$.

(iii) By Remark 4.3, $\tilde\eta_n$ satisfies (5.1), so (i) applies to σ_n. By Theorem C(iii), $\tilde\eta_n \to \bar\eta_h$, hence $\sigma_n \to \bar\sigma_h$ uniformly, and $D(\sigma_n) \to D(\bar\sigma_h)$ by continuity of G_T^*. □

Corollary 5.5 (A guaranteed bracket)**.** *Under the assumptions of Proposition 5.1(i) and Theorem D, for every $v_h \in \mathrm{P1}_{h,\psi}$ (e.g. any P1 iterate) and every $\sigma \in \mathrm{RT}_0(\mathcal{T}_h')$ with $\operatorname{div}\sigma = -f$ on any triangulation $\mathcal{T}_h'$ satisfying (4.1) (e.g. σ_n from any CR iterate),*

$$D(\sigma) \;\leq\; m_{\mathcal{W}} \;\leq\; m_{\mathcal{H}} \;\leq\; J(v_h), \qquad \textit{hence} \qquad 0 \leq m_{\mathcal{H}} - m_{\mathcal{W}} \leq J(v_h) - D(\sigma).$$

If moreover $f = 0$ and the hypotheses of [8, Thm. 1] *and of Proposition 5.1(ii) hold, then*

$$\Delta_h := \min_{\mathrm{P1}_{h,\psi}} J_h - \min_{\mathrm{CR}_{h,\psi}} J_h \;\to\; m_{\mathcal{H}} - m_{\mathcal{W}} \qquad (h\to 0).$$

Proof. The chain is Proposition 5.1(i) and Theorem D(i). For the limit, $\min_{\mathrm{P1}_{h,\psi}} J_h \to m_{\mathcal{H}}$ by Proposition 5.1(ii) and $\min_{\mathrm{CR}_{h,\psi}} J_h \to m_{\mathcal{W}}$ by [8, Thm. 1]. □

Remark 5.6 (A computable gap indicator: precise status). (a) *Proved:* the width $J(v_h) - D(\sigma)$ is, on every mesh and for every pair of iterates, a rigorous upper bound for the gap $m_{\mathcal{H}} - m_{\mathcal{W}}$ (up to floating-point arithmetic and the round-off of the linear solves, which only affects the equilibration of σ_n; its normal jumps can be monitored). If the width is below ε, the gap is below ε: *absence of a gap can be certified to any computed accuracy.* (b) *Proved, under the hypotheses of Corollary 5.5:* $\Delta_h \to m_{\mathcal{H}} - m_{\mathcal{W}}$, so $\Delta_h \to 0$ if and only if there is no gap. (c) *Not available:* a positive lower bound for the gap. It would need an upper bound for $m_{\mathcal{W}}$ or a lower bound for $m_{\mathcal{H}}$; the dual problem characterises $m_{\mathcal{W}}$, not $m_{\mathcal{H}}$ [8, Lemma 6]. The presence of a gap is therefore only *inferred* from the behaviour of Δ_h under refinement; no rate for Δ_h is proved, and extrapolations are heuristic. (d) *Variants.* If (4.1) fails but $\gamma_T \leq G(x,\cdot)$ on T (a *minorant* sampling, e.g. a replaced by $\min_T a$ for $t^p/p + a(x)t^q/q$), then $G^*(x,\cdot) \leq \gamma_T^*$ and Theorem D(i) holds with G_T^* replaced by γ_T^*; for $t^p/p + a(x)t^q/q$ with a replaced by $\min_T a$, the CR scheme with integrand γ_T is a minimum quadrature in the sense of [8, §4]; for variable exponents, sampling at $\operatorname{argmin}_T p$ is a minorant only for large t, and a valid choice is $\gamma_T^*(s) = \max_{p\in[\min_T p, \max_T p]} s^{p'}/p'$. If the CR boundary values are not the exact edge means, $D(\sigma_n)$ computed with the exact $\int_E \psi$ remains a valid lower bound. For $f \neq 0$, $x \mapsto G_T^*(|\sigma(x)|)$ is convex on T, so replacing $\int_T G_T^*(|\sigma|)$ by the vertex rule $\frac{|T|}{3}\sum_i G_T^*(|\sigma(a_i)|)$ keeps the bound rigorous; but then the CR energy itself is *not* shown to be a lower bound. (e) The CR energy lies *below* $m_{\mathcal{W}}$ also without a gap (Theorem D(ii)); "CR has the lower energy" is never by itself evidence of a gap.

5.4. **A second diagnostic: the size of discrete saddle points.** The bracket measures energies. The proof of Theorem C suggests a second computable quantity: the size of the discrete saddle point, which is also the initial value of the Lyapunov functional. Its behaviour under refinement detects both a failure of (S) and a gap.

Theorem 5.7 (Discrete diagnostics for (S) and the gap)**.** *Let $\{P_j\}$ be a fixed polygonal partition of the polygon Ω, $G(x,t) = G_j(t)$ on P_j, $(\mathcal{T}_h)$ shape-regular triangulations resolving the partition with $h \to 0$, $f \in L^\infty$ and exact loads $\ell_h = \ell$, and Lipschitz boundary data ψ with piecewise affine trace. Let $(u_h, \nabla u_h, \eta_h)$ be P1 discrete saddle points and $(u_h^{\mathrm{CR}}, \nabla_h u_h^{\mathrm{CR}}, \eta_h^{\mathrm{CR}})$ CR discrete saddle points (Theorem C), and let m_{H^1} be as in Section 3.3, with the datum ψ.*

(a) *If $\liminf_h(\|\nabla u_h\| + \|\eta_h\|) < \infty$, then* (S) *holds and $m_{\mathcal{H}} = m_{H^1}$.*
(b) *If $c|\xi|^{p_-} - c_0 \le \Phi(x,\xi) \le C|\xi|^{p_+} + c_0$ with $1 < p_- < p_+ < \infty$ and $\liminf_h(\|\nabla_h u_h^{\mathrm{CR}}\| + \|\eta_h^{\mathrm{CR}}\|) < \infty$, then* (S) *holds and $m_{H^1} = m_{\mathcal{W}}$.*
(c) *Hence: if* (S) *fails, the P1 family is unbounded in $L^2 \times L^2$, and so is the CR family under the growth assumption of (b); if $m_{\mathcal{H}} > m_{\mathcal{W}}$ and the growth assumption of (b) holds, at least one is; if moreover $G(x,t) \ge ct^2 - A$, the P1 family is.*
(d) *If* (L) *holds with $\operatorname{ess\,inf}_x g(x,1) > 0$ and $m_{\mathcal{H}} > m_{\mathcal{W}}$, then $\|\eta_h\|_{L^\infty} \to \infty$ for the P1 family.*

Proof. We write the proof for $\psi = 0$. For Lipschitz ψ with piecewise affine trace, write $u_h \in I_h\psi + V_h$; then $(\eta_h, \nabla u_h) = \ell(u_h - I_h\psi) + (\eta_h, \nabla I_h\psi)$ and $J(u_h) = (\eta_h, \nabla I_h\psi) - \ell(I_h\psi) - \int \Phi^*(x,\eta_h)$, with $\nabla I_h\psi \to \nabla\psi$ in L^2, and the argument below goes through verbatim (for CR use edge means of ψ). (a) Extract $u_h \rightharpoonup u$ in H^1_0, $\eta_h \rightharpoonup \eta$ in L^2. For $w \in C_c^\infty$, $\nabla I_h w \to \nabla w$ uniformly, so $(\eta, \nabla w) = \lim(\eta_h, \nabla I_h w) = \lim \ell(I_h w) = \ell(w)$, and by density for $w \in H^1_0$. For $q \in L^2$ with $F(q) < \infty$ let $\Pi_h q$ be the elementwise mean; G is constant in x on each element, so Jensen gives $F(\Pi_h q) \le F(q)$, and the discrete subgradient inequality together with $(\eta_h, \Pi_h q) = (\eta_h, q)$ and $(\eta_h, \nabla u_h) = \ell(u_h)$ gives $F(q) \ge F(\nabla u_h) + (\eta_h, q) - \ell(u_h)$. In the limit (weak l.s.c. of F, $\ell(u_h) \to \ell(u) = (\eta, \nabla u)$), $F(q) \ge F(\nabla u) + (\eta, q - \nabla u)$: $\eta \in \partial F(\nabla u)$, and (S) holds. Next, by the elementwise Fenchel equality, $J(u_h) = -\int \Phi^*(x,\eta_h) = D_0(\eta_h)$, and $J(u_h) \ge m_{\mathcal{H}}$ (Proposition 5.1(i)). By weak l.s.c. of $\int \Phi^*$, $m_{\mathcal{H}} \le \limsup D_0(\eta_h) \le D_0(\eta) = J(u) = m_{H^1}$ (Proposition 3.7(i)). (b) By [53, Thm. 4.2] (a compactness result for CR functions, independent of the integrand), a subsequence $u_h^{\mathrm{CR}} \to u$ in L^1 with $\nabla_h u_h^{\mathrm{CR}} \rightharpoonup \nabla u$ and $u \in W^{1,1}_0$; as $\nabla u \in L^2$, $u \in H^1_0$. The argument of (a), with the CR interpolant ($\nabla_h I^{\mathrm{CR}} w = \Pi_h \nabla w \to \nabla w$ uniformly), gives (S). By weak l.s.c. and [8, Thm. 1] (assumption (B) holds since $\phi_h = \phi$), $J(u) \le \liminf J_h(u_h^{\mathrm{CR}}) = m_{\mathcal{W}}$; as $u \in \mathcal{W}$, $J(u) = m_{\mathcal{W}}$, so $m_{H^1} = m_{\mathcal{W}}$. (c) follows from (a), (b) and Proposition 3.7(iii). (d) If $\|\eta_h\|_{L^\infty} \le C$ along a subsequence, then, since $|\eta_h| = g(x, |\nabla u_h|) \ge g(x,1)\min\{|\nabla u_h|^\delta, |\nabla u_h|^{g_0}\}$ on each element, $\|\nabla u_h\|_{L^\infty} \le C'$. By (a), (S) holds at a weak-$*$ limit (u,η) with $u \in W^{1,\infty}$, $\eta \in L^\infty$, and Proposition 3.10 gives $J(u) = m_{\mathcal{W}}$ with u Lipschitz: no gap. □

With $\nu_0 = \eta_1 = 0$, $\|\eta_h\|^2/(2r) + r\|\nabla u_h\|^2/2$ is the initial value E_0 of the Lyapunov functional of Theorem C, so (c) says that E_0 blows up under refinement for both discretisations when (S) fails (for CR under the growth assumption of (b)), and for at least one of them (for P1 if $G(x,t) \ge ct^2 - A$) when there is a gap. Unlike the bracket, Theorem 5.7 gives no quantitative information, and boundedness can only be observed, not certified, on a finite sequence of meshes.

6. Numerical experiments

6.1. **Implementation and protocol.** All computations use the Python package `glaplace` (P1 or CR elements, piecewise constant ν, η, one sparse LU factorisation per problem). The pointwise step (2.8) is solved by Newton's method in $\tau = \log t$, which converges monotonically from an upper bound ($\tau \mapsto g(e^\tau) + re^\tau$ is convex and increasing), with a bisection fallback; it agrees with bisection to a relative 10^{-13} on $2 \cdot 10^5$ random cases. All runs start from $\nu_0 = \eta_1 = 0$ and stop when *both* the scaled primal residual $\|\nabla_h u_n - \nu_n\|_h/(1 + \|\nabla_h u_n\|_h)$ and the discrete duality gap $J_h(u_n) - D_h(\tilde\eta_n)$ (with the equilibrated field (4.3)) are below tolerances; for CR with $f = 0$, $D_h(\tilde\eta_n)$ is exactly the certificate $D(\sigma_n)$ of Theorem D(iii), for P1 it is only the discrete dual value. All results were produced with one version of the code (source hash `ec5ad1de5d43`); scripts and raw data are distributed with it. Unless stated otherwise, errors are true finite element errors computed

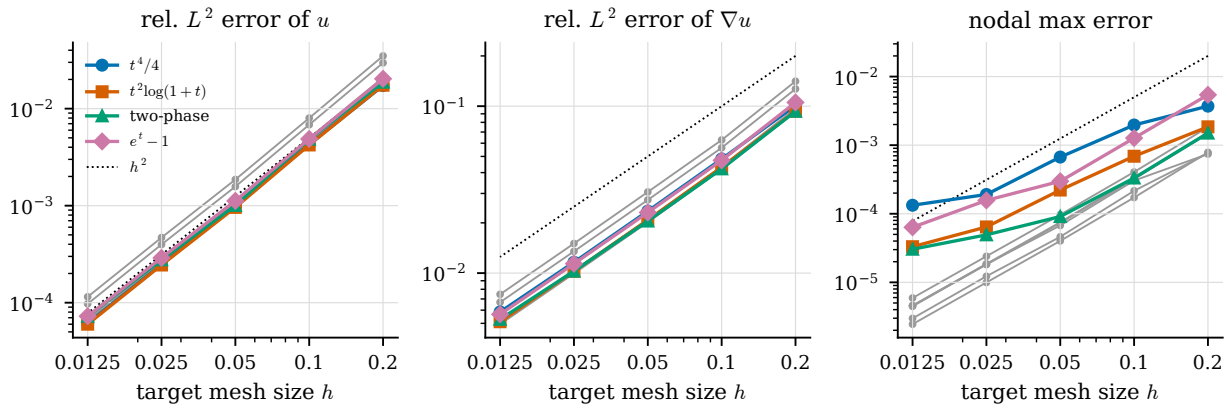


FIGURE 1. Radial torsion on the unit disk with exact solutions (P1, $r = 1$): relative L^2 errors of u (left) and ∇u (centre) and nodal maximum error (right) versus the target mesh size. Gray: $t^{1.5}/1.5$, $t^2/2$, $t^2/2+t^4/4$, $t^{1.5}/1.5+t^3/3$, $t^3/\log(1+t)$, e^t-1-t; coloured: the four cases with reduced maximum-norm order ($t^4/4$, $t^2\log(1+t)$, two-phase with unresolved interface, and e^t-1 with a dead core). Dotted: reference slopes.

by high-order quadrature against exact solutions, and orders are least-squares fits over the three finest meshes. *All orders, iteration counts and extrapolations below are observations; no rate is proved.*

6.2. **Smooth problems: orders, the parameter r, and the Lyapunov functional.** *Radial torsion.* On the unit disk with $f = c$ and zero data, the exact solutions of Remark 3.3 are available for any radial integrand. We use unstructured meshes with target sizes $h = 0.2, \dots, 0.0125$ (88 to 23 227 nodes), $c = 1$, $r = 1$ and tolerance 10^{-9}, except $e^t - 1$ ($c = 8$, $r = 30$), whose solution has a dead core $\{|x| \le 1/4\}$. Table 2 and Figure 1 show the results. The observed orders are 2 in $L^2(u)$ and 1 in $L^2(\nabla u)$ (gradient order 1.25 for the discontinuous radial exponent, 1.90/0.98 for the two-phase case with unresolved interface), including $e^t - 1 - t$ (no Δ_2), $t^{1.5}/1.5$ (not covered by Glowinski's theorem) and variable exponents with $p^- < 2$, continuous or discontinuous across the circle $|x| = 1/2$ (resolved by the mesh). The nodal maximum error has reduced order (1.1–1.4) for $t^4/4$, $t^2\log(1+t)$ and $e^t - 1$, which we attribute, without proof, to the limited smoothness of u at the origin, and for the two-phase coefficient $\mathbf{1}_{|x|>1/2}$ when its interface is not resolved by the mesh. For $e^t - 1$ the two finest meshes did not reach the residual tolerance in 20 000 iterations, although the duality gap was below 10^{-9}; the computed solution is flat in the dead core ($|\nabla u_h| \le 2.2 \times 10^{-8}$ for $|x| < 0.2$). For t^p/p with $p = 1.2$ and $p = 1.1$ none of the eight runs reached the residual tolerance in 50 000 iterations (final residuals up to 2×10^{-8} and 4×10^{-7}), with duality gaps below 8×10^{-9}; the iteration slows down as $p \to 1$, where ∂F becomes multivalued in the limit. For the radial variable exponents ($p^- < 2$) the discrete diagnostics $\|\nabla_h u_h\|$, $\|\eta_h\|$ and $\max|\eta_h|$ converge under refinement, consistently with (S) (Theorem 5.7 gives only the converse implication), and on the finest mesh the modular $\int |\nu_n - \nabla u_h|^{p_T}$ decays to 10^{-12}–10^{-16} in 1000 iterations, in line with Corollary 2.15, which concerns the continuous iteration; CR elements give the same orders with slightly smaller errors and fewer iterations.

Manufactured solution. On $(0,1)^2$ with $G(x,t) = t^2/2 + (1+x_1)t^4/4$, $u = e^{x_1+x_2}$, Dirichlet data u, $r = 100$ and tolerance 10^{-10}, the pairwise orders are 2.02, 2.00, 2.00 (L^2) and 1.00 (gradient).

The parameter r. Theorem C guarantees convergence for every $r > 0$; the speed depends strongly on r (Figure 2). On the checkerboard problems of Section 6.4 the iteration count varies by a factor of about 14 (P1) to 38 (CR) over $r \in [0.1, 100]$, with the best r near 1 for CR and between 1 and 3

TABLE 2. Smooth problems with exact solutions (P1). Fitted orders of the relative L^2 errors of u and ∇u and of the nodal maximum error over the three finest meshes, and iterations on the coarsest and finest mesh. Top: radial torsion on the unit disk, $h = 0.2, \dots, 0.0125$; middle: $p \to 1$, $h = 0.2, \dots, 0.025$; radial variable exponents ($|x| = 1/2$ resolved, $h = 0.2, \dots, 0.0125$); bottom: manufactured solution on $(0,1)^2$, $n = 10, \dots, 80$, pairwise orders. Column "theory": results that apply to the *continuous* iteration, given that (S) holds for these data (Remark 3.3); "G": Glowinski's theorem via Lemma 2.8; "B": Theorem B or Corollary 2.15. The discrete iteration is covered by Theorem C in all cases. *: residual tolerance not reached on the finest mesh(es).

G	theory	$L^2(u)$	$L^2(\nabla u)$	max	iterations
$t^{1.5}/1.5$	B	2.01	1.02	1.94	147/837
$t^2/2$	G, B	2.00	1.02	2.02	29/29
$t^4/4$	G, B	2.00	1.00	1.17	47/176
$t^2/2 + t^4/4$	G, B	2.00	1.01	2.02	30/26
$t^{1.5}/1.5 + t^3/3$	G, B	2.01	1.02	1.98	148/843
$t^2 \log(1+t)$	G, B	2.00	1.01	1.37	36/60
$t^3/\log(1+t)$	G, B	2.00	1.02	2.02	41/33
$e^t - 1 - t$	G, B(b)	2.00	1.01	2.02	29/25
$t^2/2 + \mathbf{1}_{\lvert x\rvert>1/2} t^4/4$ (unresolved)	G, B	1.90	0.98	0.79	30/27
$e^t - 1$, $c = 8$, $r = 30$	G	1.97	1.01	1.11	252/20000*
$t^{1.2}/1.2$	B	1.98	1.03	–	50000*
$t^{1.1}/1.1$	B	1.84	1.02	–	50000*
$t^{p(x)}/p(x)$, $p = 1.4 + 0.4\lvert x\rvert$	B	2.03	1.01	–	195/3733
$t^{p(x)}/p(x)$, $p = 1.5 + \lvert x\rvert$	B	2.03	1.02	–	59/569
$t^{p(x)}/p(x)$, $p = 1.5$ for $\lvert x\rvert < \frac{1}{2}$, 3 else	B	2.03	1.25	–	125/623
$t^2/2 + (1+x_1)t^4/4$, $u = e^{x_1+x_2}$	G, B	2.00	1.00	–	163/218

for P1, and grows moderately under refinement at fixed r. For the manufactured problem ($n = 40$) the counts for $r = 1, 10, 30, 100, 300$ are 11 575, 1167, 396, 208, 516, with equal L^2 errors to four digits. For the dead core problem ($e^t - 1$, $h = 0.05$, tolerance 10^{-8}) the counts for $r = 1, 3, 10, 30, 100$ are 6403, 2139, 650, 244, 529, but for $r = 100$ the error at termination is 1.1292×10^{-3} against 1.1204×10^{-3} for $r \le 10$: for large r a small primal residual does not mean a small iteration error. The duality gap (1.3×10^{-8} against 2.1×10^{-9}) and the dual residual flag this run; this is why we stop on both residual and gap.

The Lyapunov functional. With a reference saddle point computed to residual 10^{-14}, E_n of (2.9) was non-increasing at every iteration in all runs checked: CR on the $p(x)$ checkerboard ($\lambda = 3$, $n = 32$, $r = 1, 3, 10$; $E_{100}/E_1 = 3 \times 10^{-14}$, 6×10^{-7}, 8×10^{-4}) and P1 on the disk ($h = 0.1$) for five integrands including the nonsmooth $e^t - 1$. The decay is geometric after a transient, with rates depending on G and r; no rate is proved.

6.3. **(S) fails, but there is no gap.** Example 3.9 can be computed: $p = 3/2$ on the unit disk with $f = |x|^{-\beta}$. For $\beta = 1.5$ the exact solution is $4\log(1/|x|)$, $\nabla u \notin L^2$ and (S) fails; for $\beta = 1.4$, $|\nabla u| \sim |x|^{-0.8} \in L^2$ and (S) holds (Proposition 3.5(i)); for $\beta = 1.6$, (S) fails again. The singular load must be integrated accurately (we use a Gauss–Jacobi rule on the triangles at the origin; the barycentre rule gives an energy error 2.5 times larger and a far too small $\|\nabla u_h\|$). Figure 3 shows the diagnostics of Theorem 5.7. For $\beta = 1.5$, $\|\nabla_h u_h\|^2$ grows by about $32\pi \log 2 \approx 69.7$ per halving of h (observed +67, +75, +61, +69), consistently with $\int_{|x|>h} |\nabla u|^2 = 32\pi \log(1/h)$, while

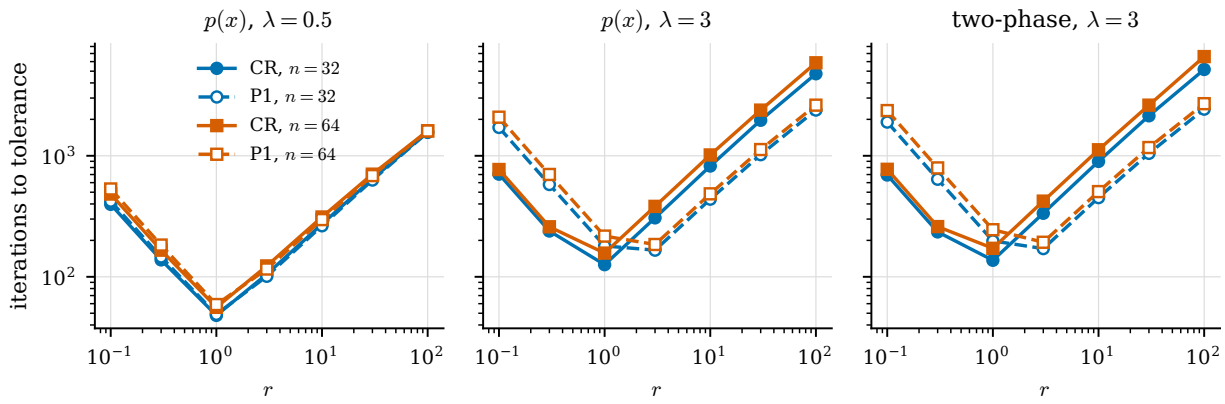


FIGURE 2. Iterations to tolerance (scaled residual $< 10^{-10}$ and duality gap $< 10^{-9}$) versus the penalty parameter r, for CR (solid) and P1 (dashed) on criss-cross meshes with $n = 32$ and $n = 64$: $p(x)$ checkerboard with $\lambda = 0.5$ (no gap) and $\lambda = 3$, and two-phase checkerboard with $\lambda = 3$ (Section 6.4). Theorem C guarantees convergence for every r; nothing is proved about the speed.

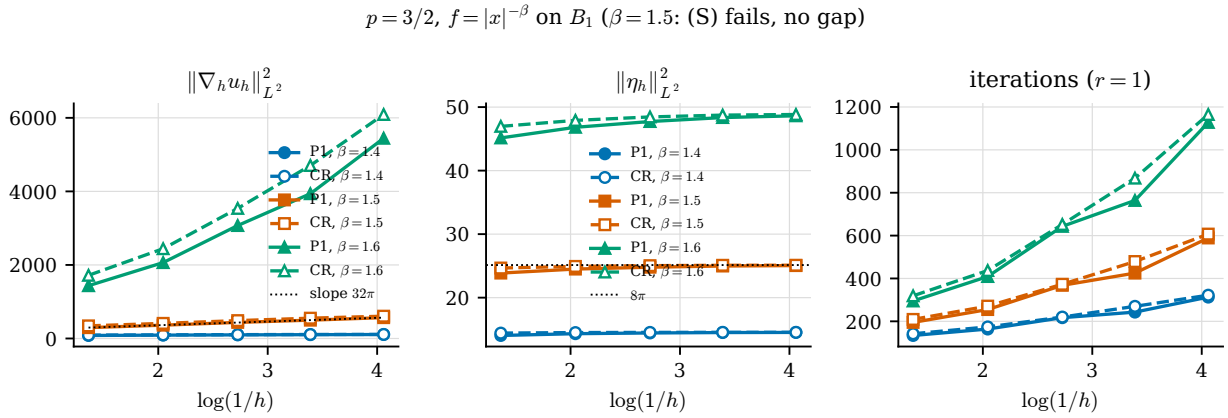


FIGURE 3. $G = t^{3/2}/(3/2)$, $f = |x|^{-\beta}$ on the unit disk, P1 (solid) and CR (dashed), $r = 1$: squared L^2 norms of the discrete gradient (left) and of the discrete flux (centre), and iterations (right), versus $\log(1/h)$. For $\beta = 1.5$ and 1.6 (S) fails and $\|\nabla_h u_h\|$ grows without bound (dotted: slope 32π predicted for $\beta = 1.5$), while the flux stays bounded (dotted: 8π); for $\beta = 1.4$ (S) holds. There is no Lavrentiev gap in any case.

$\|\eta_h\|^2 \to 8\pi = \|\eta\|^2$ stays bounded, and the iteration counts grow (195 → 589). For $\beta = 1.4$ the increments of $\|\nabla_h u_h\|^2$ decrease. The P1 and CR energies approach the same limit $J(u) = -32\pi/3$, P1 from above (since $u_h \in W_0^{1,3/2}$; up to the quadrature of the singular load), slowly ($J(u_h) - J(u) = 7.58 \to 1.97$); the CR energies are also above $J(u)$ here; this does not contradict Theorem D, which does not apply ($\Omega_h \neq \Omega$, f not piecewise constant, and the statement on the CR energy itself needs $f = 0$). Thus failure of (S) is visible in the iteration and in the size of the discrete saddle points, but it is not a consistency problem: P1 is adequate.

6.4. **The Lavrentiev experiments: setting.** We take $\Omega = (-1, 1)^2$, $f = 0$, and the boundary datum λu_0, where $u_0 = 1$ on $\{|x_1| < x_2\}$, $u_0 = -1$ on $\{|x_1| < -x_2\}$ and $u_0 = x_2/|x_1|$ elsewhere; its

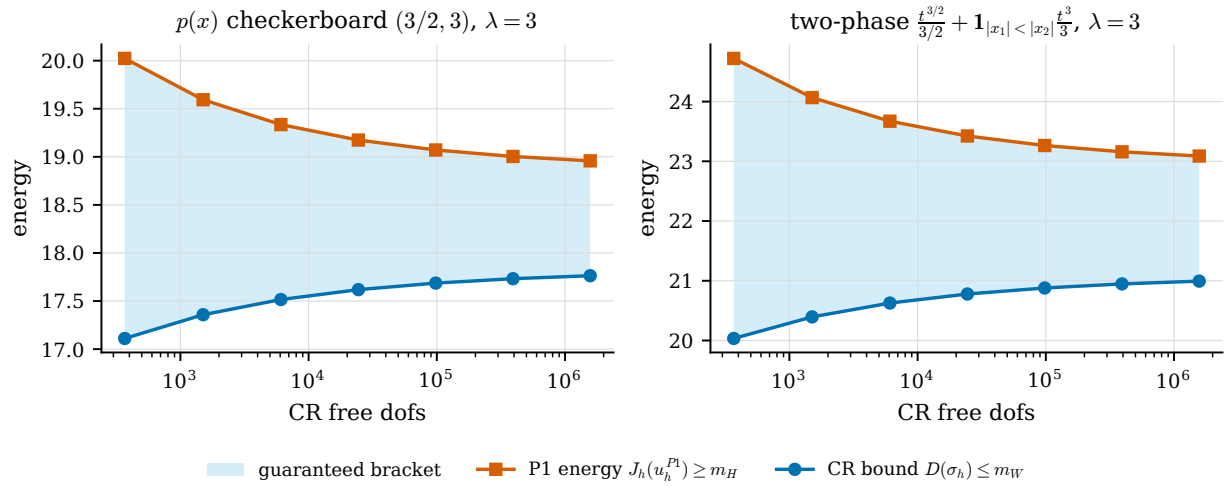


FIGURE 4. Guaranteed brackets $D(\sigma_h) \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq J_h(u_h^{\mathrm{P1}})$ (shaded) versus the number of CR unknowns, for the $p(x)$ checkerboard (left) and the two-phase checkerboard (right), boundary datum $3u_0$, criss-cross meshes $n = 8, \ldots, 512$. Upper curve: P1 energy (Proposition 5.1); lower curve: CR energy $= D(\sigma_h)$ (Theorem D). Both curves move slowly; their difference does not tend to 0 at any visible rate.

trace is λx_2. The integrands are the Zhikov-type checkerboard exponent [57]

$$G(x,t) = t^{p(x)}/p(x), \qquad p = 3 \text{ on } \{|x_1| < |x_2|\}, \quad p = \tfrac{3}{2} \text{ elsewhere},$$

and the two-phase checkerboard $G(x,t) = t^{3/2}/(3/2) + \mathbf{1}_{\{|x_1|<|x_2|\}}t^3/3$. These are Experiments 1 and 3 of Balci, Ortner and Storn [8], who report, from numerical evidence, a gap for $\lambda > 1$ and for $\lambda \geq 0.4$ respectively; for $\lambda = 1$ and the $p(x)$ problem the minimiser is x_2 with energy 2. We use criss-cross meshes ($n \times n$ squares, each cut into four triangles by its diagonals, $n = 8, \ldots, 512$), which resolve $|x_1| = |x_2|$, so that (4.1) holds; P1 nodal data and CR edge means are exact, $r = 3$, and the tolerances are 10^{-10} (residual) and 10^{-9} (relative duality gap). All 296 runs converged; the largest CR problem ($n = 512$, 1.57×10^6 unknowns) took 792 iterations. By Proposition 5.1(i), Theorem D(ii) and Remark 5.3, all assumptions of Corollary 5.5 hold: every computed P1 energy is an upper bound for $m_{\mathcal{H}}$, every CR energy a lower bound for $m_{\mathcal{W}}$, and $\Delta_h \to m_{\mathcal{H}} - m_{\mathcal{W}}$. The certificate is exact up to round-off: the normal jumps of σ_n never exceeded 1.1×10^{-13}, and $J_h(u_n) - D(\sigma_n)$ was never below -8×10^{-10} (relative 4×10^{-11}), so the bounds hold up to an absolute error of order 10^{-10}. The conjugates of the two-phase integrand, which have no closed form, are evaluated as rigorous upper bounds ($s\,t_{\mathrm{hi}} - G(t_{\mathrm{lo}})$ on a verified bracket), so their evaluation does not affect the direction of the bound. For $\lambda = 1$ both energies equal 2 to within 1.5×10^{-11} on every mesh.

6.5. **Brackets, widths and the signature of a gap.** Figure 4 shows the bracket for $\lambda = 3$ and Table 3 the finest-mesh values for a range of λ. The rigorous conclusions are, for instance,

$$17.7635 \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq 18.9581 \quad (p(x),\ \lambda = 3),$$
$$20.9938 \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq 23.0881 \quad (\text{two-phase},\ \lambda = 3),$$

so the gap is at most 1.195 and 2.094, respectively; and, at the other end, the gap is at most 8.0×10^{-7} for $p(x)$ with $\lambda = 0.5$ and at most 2.5×10^{-8} for the two-phase problem with $\lambda = 0.2$. These are certificates of (near) absence of a gap; positivity of a gap cannot be certified (Remark 5.6(c)).

TABLE 3. Finest-mesh brackets for the checkerboard problems: P1 energy ($\geq m_{\mathcal{H}}$), best certified lower bound $D(\sigma_n)$ ($\leq m_{\mathcal{W}}$, equal to the CR energy up to 10^{-10}), width $\Delta_h \geq m_{\mathcal{H}} - m_{\mathcal{W}}$, ratio $\Delta_{n/2}/\Delta_n$ on the two finest meshes, and least-squares slope of Δ_h in the number of unknowns over the four finest meshes. A ratio 4 (slope -1) is $O(h^2)$ closure.

problem, λ	n	P1 energy	lower bound D	width Δ_h	ratio	slope
$p(x)$, 0.5	512	0.51381346	0.51381265	8.0×10^{-7}	4.00	−1.00
$p(x)$, 1	256	2.00000000	2.00000000	$\leq 1.5 \times 10^{-11}$	–	–
$p(x)$, 1.08	256	2.33326877	2.33325614	1.3×10^{-5}	2.02	−0.53
$p(x)$, 1.5	256	4.52197423	4.51556823	6.4×10^{-3}	1.31	−0.24
$p(x)$, 2	256	8.12513230	8.02915903	9.6×10^{-2}	1.17	−0.15
$p(x)$, 3	512	18.95812228	17.76348924	1.195	1.06	−0.06
$p(x)$, 4	256	35.76099252	30.51017011	5.251	1.07	−0.06
two-phase, 0.2	512	0.24363748	0.24363745	2.5×10^{-8}	3.23	−0.84
two-phase, 0.4	256	0.71309620	0.71309059	5.6×10^{-6}	2.45	−0.67
two-phase, 1	256	3.15984497	3.15462093	5.2×10^{-3}	1.34	−0.26
two-phase, 3	512	23.08812916	20.99380420	2.094	1.06	−0.06
two-phase, 4	256	41.75864138	34.43734359	7.321	1.06	−0.06

Figure 5 shows Δ_h versus the number of unknowns for all λ. Two regimes are clearly separated. For small λ the width closes at a steady rate; for large λ the closure ratio per refinement decreases towards 1 ($1.30, 1.23, 1.17, 1.12, 1.09, 1.06$ for $p(x)$ with $\lambda = 3$), i.e. the width stalls. By Corollary 5.5, Δ_h converges to the gap; a width that stalls at a value of order 1 is the signature of a gap, although it is not a proof of one. Heuristic extrapolations (Aitken, or least-squares fits with a free exponent, over the three or four finest meshes) give closure exponents ≈ 0.56–0.60 in h that *decrease* with refinement, and estimated gaps of about 1.04 ($p(x)$, $\lambda = 3$) and 1.85 (two-phase, $\lambda = 3$) from the free-exponent fits (fixed exponents 1 or 2 would give up to 1.17 and 2.06); because the fitted exponent keeps drifting, we regard these numbers only as conjectural, the rigorous statement being the bracket.

No-gap controls, and a first surprise. For the $p(x)$ problem with $\lambda \in \{0.25, 0.5\}$ the ratios tend to 4.00 (slope -1.00): both bounds close at $O(h^2)$, as expected without a gap (cf. the $O(\mathrm{ndof}^{-1})$ reported in [8, Exp. 1]); for $\lambda = 0.75$ the ratios ($3.0 \to 3.84$) are still pre-asymptotic. For the two-phase problem with $\lambda \in \{0.1, 0.2\}$, however, the ratios are only 3.2–3.5 (slope ≈ -0.85), also on $n = 512$, with both bounds closing at the same reduced rate. We expected 4; we have no explanation. It is not a solver artefact: the final discrete duality gaps of these runs are below 3×10^{-12}, four orders of magnitude below the widths, which are in any case below 2.5×10^{-8}. So the rate of closure alone does not separate gap from no gap; its trend under refinement did in all our runs (an observation, not a theorem).

Near the thresholds, a second surprise. [8] indicate, numerically, a gap for $p(x)$ with $\lambda > 1$ and for the two-phase problem with $\lambda \geq 0.4$. Our widths for $\lambda \in (1, 1.08]$ ($p(x)$) and $\lambda \in \{0.4, 0.5, 0.6\}$ (two-phase) still decrease steadily on uniform meshes up to $n = 256$ (ratios 2.0–2.5, slowly declining), so they neither confirm nor exclude a gap there; rigorously, the gap is at most 1.3×10^{-5} for $p(x)$ with $\lambda = 1.08$ and at most 1.2×10^{-4} for the two-phase problem with $\lambda = 0.6$. Figure 6 compares our widths with the differences $F(u_c) - F_h(u_{nc})$ of [8, Fig. 1], read from the vector graphics (the colour of each curve is identified from the legend samples). Their adaptive meshes reach values 10–100 times smaller than our uniform meshes. Since their initial meshes resolve the exponent, their conforming energies are exact and their CR energies have $f = 0$, Proposition 5.1 and Theorem D make their differences upper bounds for the gap as well, provided their discrete CR problems are

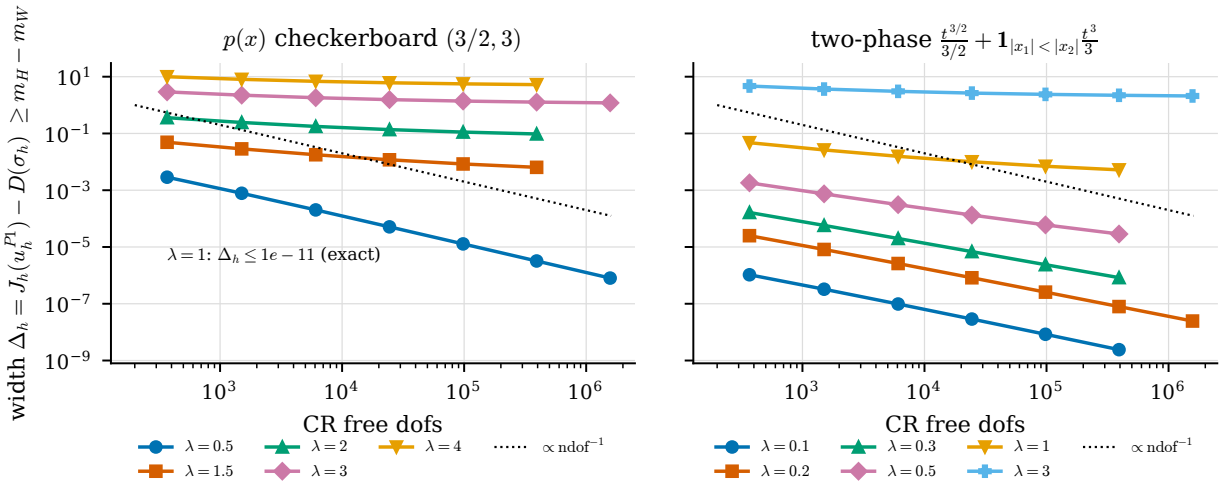


FIGURE 5. Width $\Delta_h = J_h(u_h^{\mathrm{P1}}) - D(\sigma_h)$ of the guaranteed bracket, a rigorous upper bound for $m_{\mathcal{H}} - m_{\mathcal{W}}$, versus the number of CR unknowns, for the $p(x)$ checkerboard (left) and the two-phase checkerboard (right) and several data λu_0. Dotted: slope ndof^{-1}, i.e. $O(h^2)$. For $p(x)$ with $\lambda = 1$ the width is at round-off level on all meshes. Small λ: steady closure; large λ: the width stalls, the signature of a Lavrentiev gap.

solved accurately; read this way, their data bound the gap at $\lambda = 1.08$ by about 4×10^{-7} (observed). Neither data set shows a plateau for $\lambda \leq 1.08$; in both, however, the closure rate *decreases* under refinement for $\lambda \geq 1.04$ (our ratios $2.28 \to 2.21$, $2.24 \to 2.11$, $2.21 \to 2.02$; their slope for $\lambda = 1.08$ drops from about -1.1 to -0.45), while it is steady or increasing for $\lambda = 0.99$, the pattern of the gap cases. Since $m_{\mathcal{H}} - m_{\mathcal{W}}$ vanishes at $\lambda = 1$, a gap below the final widths is compatible with all the data. We conclude only that the gap for $\lambda \in (1, 1.08]$, if present, is below 1.3×10^{-5} (our bound) and plausibly below 4×10^{-7}; it cannot be certified either way. The widths stall, the signature of a gap (not a proof, Remark 5.6(c)), for $\lambda \geq 1.5$ ($p(x)$) and $\lambda \geq 1$ (two-phase).

Quadrature. Reproducing [8, §4.2] ($p(x)$, $\lambda = 5$), replacing the exponent on the nodal patch of the origin by its maximum sends *both* P1 and CR towards the P1 value (59.73 for exact sampling on $n = 256$; perturbed: P1 60.66, CR 59.91), and replacing it by its minimum sends both towards the CR value (45.93; perturbed: 45.85, 45.52). The perturbed energies are not bounds for the original problem: the guarantees of Proposition 5.1 and Theorem D depend on the sampling (majorant for P1, minorant for CR, Remark 5.6(d)).

6.6. **The size of the discrete saddle points.** Figure 7 shows the quantities of Theorem 5.7. In the gap regime $\max|\eta_h^{\mathrm{P1}}|$ grows like $h^{-0.9}$ (e.g. $104.8 \to 367.4 \to 706.4$ for $p(x)$, $\lambda = 3$, $n = 64, 256, 512$; $138 \to 492 \to 951$ for the two-phase problem), consistent with Theorem 5.7(d) (the gap for these data is inferred, not proved); the CR flux grows much more slowly ($11.9 \to 22.7 \to 31.6$). A flux growing like $h^{-0.9}$ at a single point is consistent with a singularity $|\eta| \sim |x|^{-0.9}$, which lies in L^2; so this growth says nothing against (S). Growth of $\max|\eta_h|$ alone is not a gap indicator: it grows mildly also without a gap ($0.54 \to 0.57 \to 0.59$ for the two-phase problem with $\lambda = 0.2$); the observed rates differ by an order of magnitude, but we have no theory for them. The CR gradient norm $\|\nabla_h u_h^{\mathrm{CR}}\|$ increases slowly ($7.57 \to 7.73 \to 7.82$), roughly like $(\log(1/h))^{1/2}$, which is consistent with, but does not establish, an unbounded CR family. The P1 norms $\|\nabla u_h\|$ ($6.565 \to 6.612 \to 6.627$) and $\|\eta_h\|$ ($9.145 \to 9.158 \to 9.212$) look bounded up to $n = 512$, except for the last increment of $\|\eta_h\|$. By Theorem 5.7(a), bounded P1 saddle points would imply (S)

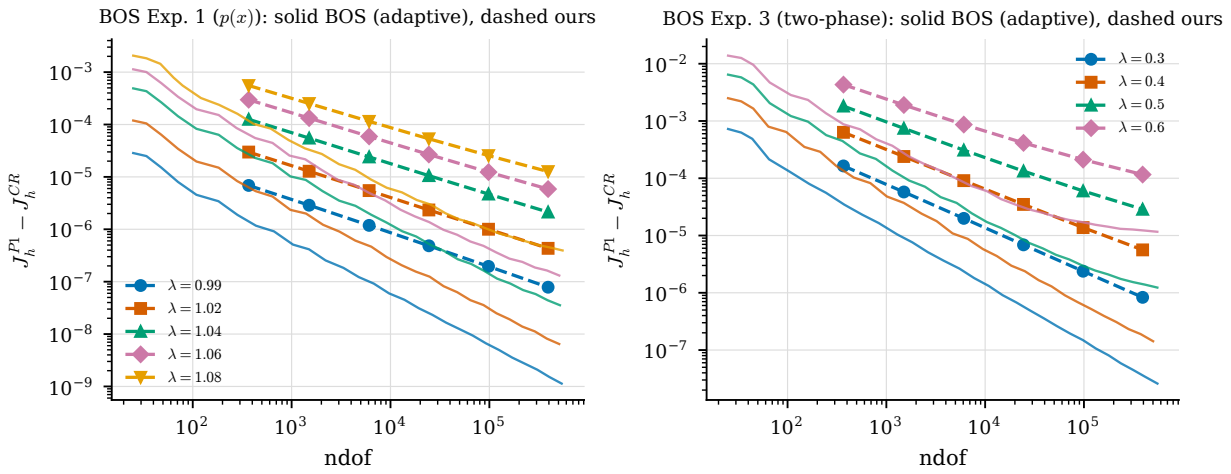


FIGURE 6. Near the gap thresholds reported in [8]: our certified widths $J_h^{\mathrm{P1}} - D(\sigma_n)$ (equal to $J_h^{\mathrm{P1}} - J_h^{\mathrm{CR}}$ to within 3×10^{-12}; uniform criss-cross meshes, ADMM; dashed, markers) and the corresponding differences digitised from [8, Fig. 1] (adaptive meshes; solid), versus the number of unknowns. Left: $p(x)$ checkerboard, $\lambda = 0.99, \dots, 1.08$; right: two-phase checkerboard, $\lambda = 0.3, \dots, 0.6$. By Corollary 5.5 our curves bound the gap from above, and so do theirs provided their CR problems are solved accurately (see text); none shows a plateau, but the closure rate decreases for $\lambda \geq 1.04$ in both.

together with $m_{\mathcal{H}} = m_{H^1}$, i.e., if the gap inferred above is real, (S) *with* a gap, in line with the discussion after Proposition 3.12; the data are too short to decide.

6.7. **Beyond the checkerboards.** *Double phase with the Esposito–Leonetti–Mingione weight.* With $G = t^{1.5}/1.5 + a(x)t^q/q$, $a = \max(x_2^2 - x_1^2, 0)/|x|$ (Lipschitz, so $\alpha = 1$), data λu_0 and $f = 0$, the no-gap range of [19, Thm. 2.3] is $q \leq 2.5$, while $q = 4 > d + \alpha$ is in the range of the counterexamples of [38]. Here a is not piecewise constant; a rigorous bracket is obtained from a majorant sampling for P1 and a minorant sampling for CR (Remarks 5.2(c) and 5.6(d)), but its width is dominated by the $O(h)$ sampling error, e.g. $[82.15, 82.85]$ for $q = 2.4$, $\lambda = 8$, $n = 256$, and $[101.6, 196.6]$ for $q = 4$, $\lambda = 8$ (gap ≤ 95.0). The difference of the barycentre-sampled P1 and CR energies, which has no theory, is more informative: for $q = 2.4$ it closes at $O(h^2)$ (ratio $\to 4$) for all $\lambda \leq 8$; for $q = 4$ it stalls for $\lambda \geq 4$ (ratios $1.59 \to 1.30$ for $\lambda = 4$), with $\max|\eta_h^{\mathrm{P1}}|$ growing, is ambiguous at $\lambda = 2$ (steady ratio ≈ 2.45) and closes at $O(h^2)$ for $\lambda \leq 1$; for $q = 2.8$, $\lambda = 8$ it closes at a ratio 2.4–2.6 that increases with n, undecided at $n \leq 128$.

An exponential checkerboard. For $G = a(x)(e^t - 1 - t) + t^2/2$ with $a = \mathbf{1}_{\{|x_1| < |x_2|\}}$, which satisfies neither the growth condition of [8] nor Δ_2, Theorem D still applies; at $n = 256$ the bracket is $[0.740687, 0.740812]$ for $\lambda = 0.5$ and $[35.18, 37.45]$ for $\lambda = 3$, and the width stalls for $\lambda \geq 2$ (ratio 1.13 at $\lambda = 3$), the pattern of the gap cases above. We know no theory for the gap in this problem.

6.8. **Discussion.** The experiments confirm the division of labour suggested by the theory. The solver is never the problem: the iteration converged in every Lavrentiev run, for every r, as Theorem C guarantees, and failure of (S) showed up only as iteration counts growing under refinement (observed for one example, Section 6.3). What a gap changes is the meaning of the computed numbers, and here the bracket of Corollary 5.5 turns two separately convergent but differently targeted schemes into a computable statement about $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$. Three observations did not match our expectations and are reported as such: the no-gap two-phase controls close at ratio 3.2–3.5

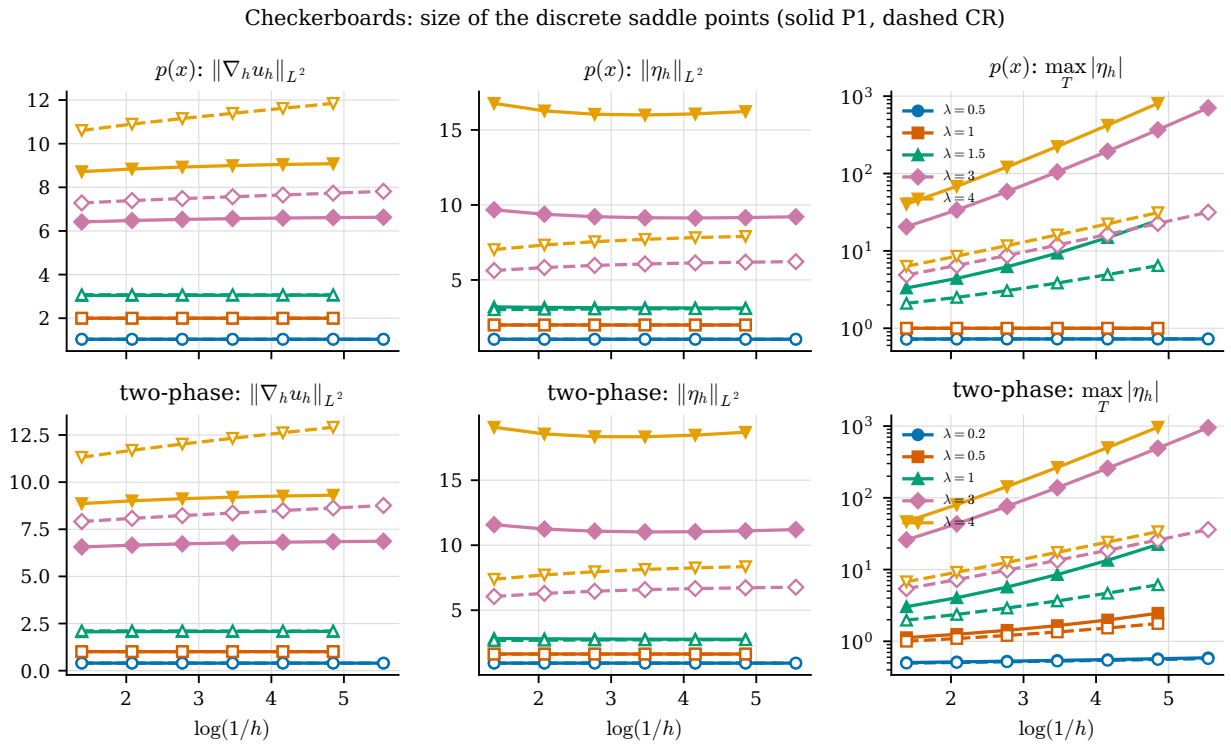


FIGURE 7. Size of the discrete saddle points for the checkerboard problems, P1 (solid) and CR (dashed), several data λu_0, versus $\log(1/h)$: $\|\nabla_h u_h\|_{L^2}$ (left), $\|\eta_h\|_{L^2}$ (centre) and $\max_T|\eta_h|$ (right, logarithmic). Top: $p(x)$ checkerboard; bottom: two-phase checkerboard. In the (numerically inferred) gap regime the maximal P1 flux blows up, consistently with Theorem 5.7(d); the L^2 norms grow slowly (CR) or not visibly (P1).

rather than 4; near the reported thresholds neither data set shows a plateau, although both show a decreasing closure rate; and the P1 saddle points in the gap regime look bounded in L^2 up to $n = 512$, which, if true, would be, if the gap is real, an instance of (S) with a gap. Limitations: widths certify upper bounds only, extrapolations are heuristic, and the disk errors are computed on inscribed polygons against quadrature-computed radial solutions.

7. Conclusions and open problems

For an arbitrary convex Young integrand and $\rho = r$, convergence of Glowinski's method at the continuous level is equivalent to boundedness of the iterates and hinges on (S); under (S) the iteration converges weakly in general and strongly under Lieberman's condition, with no continuity in x. (S) is a regularity property that is not available in general for double phase, two-phase and exponential energies, and it is a different property from the absence of a Lavrentiev gap. At the discrete level the iteration converges unconditionally, for P1 and CR alike; the remaining issue is consistency, which is governed by the gap. Under the assumptions of [8, Thm. 1] (polynomial growth, resolved or minimum quadrature), CR elements give the method a convergence theory in the Lavrentiev regime (Theorem C and [8, Thm. 1]), and the CR iterates carry a guaranteed lower bound for $m_{\mathcal{W}}$ which, together with the P1 energy, brackets the gap.

Open problems: (1) the behaviour of u_n when (S) fails (then (η_n, ν_n) is unbounded; cf. [45, Rem. 23]); (2) strong convergence of u_n for general convex G, the case $\rho \neq r$ without Glowinski's

(iii)–(iv), and rates for the iteration, including their dependence on h and r and a theory for the choice of r; (3) whether (S) can hold together with a gap for loads $f \in L^d$ (Example 3.8 is degenerate; the P1 data of Section 6.6 are suggestive but not conclusive); (4) a computable *lower* bound for $m_{\mathcal{H}}$, which would certify the positivity of a gap; (5) rates for the width Δ_h, the observed $O(h^2)$ closure without a gap and its reduced rate for the two-phase controls, and the extension of Theorem D to $f \neq 0$ for the CR energy itself and to non-resolved coefficients with sharper than $O(h)$ sampling errors; (6) global regularity, hence (S), for Musielak–Orlicz $G(x,t)$ without Δ_2 and for double phase energies up to the boundary.

Software availability. The Python package `glaplace` (NumPy/SciPy; MIT license), which implements the method with P1 and CR elements for arbitrary Young functions on two-dimensional triangulations, including the certified lower bound of Theorem D, and contains the scripts and raw data of all experiments of Section 6, is available at `https://github.com/jfbonder/glaplace`. It generalises the code for the $p(x)$-Laplacian of [4].

Declaration of the use of AI tools. Generative AI was used extensively in this work, in the form of Claude models developed by Anthropic (used through the Claude desktop application and its agent tools, October 2026). The research problem, the choice of method and of the test problems, and the scientific direction are the authors'. The AI tools were used for: (i) writing the Python code of the solver and of all numerical experiments, including the tests, starting from the authors' original MATLAB code for the $p(x)$-Laplacian (the code of [4]), which was translated to Python and extended to general Young functions and to Crouzeix–Raviart elements; (ii) drafting the proofs of Theorems A–D and of the auxiliary results, starting from the authors' earlier unpublished argument at the continuous level; (iii) searching and compiling the literature, extracting the hypotheses of the cited results from the sources, and checking the bibliographic data; (iv) drafting and editing the text of the article; (v) independent critical reviews of the mathematics and of the exposition, carried out by separate instances of the model; and (vi) in the final rewriting, a team of AI agents with separate roles (numerical analyst, expert in nonlinear partial differential equations, programmer, bibliography reviewer, editor and translator), whose drafts, computations and reports were checked against each other, followed by two further independent referee reports by AI agents (numerical analysis and PDE), which were applied to the text. The numerical results reported here were produced by running the code described above. The authors take full responsibility for the content of the article.

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